

# An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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## ABSTRACT

Searches for narrowband transmitters above about 30 GHz have covered a single spectral line toward some tens of nearby stars. Uniformly searched, 404 ALMA execution blocks toward 60 systems give 402 fine-channel windows with 15.3–1953 kHz channels, covering 47.7 GHz of union bandwidth in 26 disjoint intervals between 114 and 873 GHz; 1249 coarser windows carry no drift discrimination. Windows are filtered for an unresolved carrier drifting linearly in the stellar frame; crossings are tested against catalogued transitions at the stellar velocity, and unattributed ones sought again in later blocks. Depth is a measured recovery fraction: for a carrier narrower than the instrumental channel, nine injections in ten return at  $1.6 \times 10^{15}$  W for the median system,  $8.7 \times 10^{13}$ – $3.7 \times 10^{16}$  W across systems. The drift ceiling is 12.0 nHz, 6 per cent of the Sheikh et al. (2019) range. Of the 56 crossings, 18 fall on a catalogued transition and 38 do not; four are in one block failing a quality criterion applied post-inspection. None of the unattributed crossings recurred, including the two that lead their own control fields, while 13 attributed to circumstellar or foreground carbon monoxide do recur, as expected. That excludes a carrier holding its stellar-frame frequency to within  $\pm 0.21 \text{ km s}^{-1}$ , radiating through both observations, not a persistent transmitter of any kind; the remainder is consistent with chance to the control positions’ accuracy. Line confusion costs 6.4 per cent of the coverage and 21 systems have no confirmation epoch; these, not sensitivity, limit the search.

*Subject headings:* extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

## 1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrowband transmitter at the star’s own position. No such signal was found.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only  $6 \times 10^{-18}$  of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below  $\sim 30$  GHz. Two targeted searches have reached nearby stars above that frequency. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz, and Mauersberger et al. (1996) included  $\tau$  Cet and  $\epsilon$  Eri among 17 targets at the same frequency; both searched a single line about 35 MHz wide at one epoch, and Mason et al. (2025) has since searched more distant stars in Band 3. This survey covers 47.7 GHz in 26 disjoint intervals, searches a grid of drift rates rather than a fixed frequency, tests every crossing against the 512 control positions of its own field, and uses repeat observations to test persistence.

The band has physical advantages and physical costs. Interstellar dispersion and scintillation are negligible at these frequencies, so a carrier arrives essentially undistorted, and a given aperture delivers far more gain per watt supplied. Against that, receiver system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the same high gain narrows the beam: the 12 m benchmark of §6.3 radiates into a beam only 34 arcsec across, so the Earth must lie inside a beam that was deliberately aimed. What else makes the band hard belongs to the archive rather than to the sky, and §2 sets that out.

The signal sought is an *unresolved spectral carrier* — emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch. Recurrence rather than brightness is what separates an interesting event from a detection (Sheikh et al. 2021), and the archive holds a second, separated epoch for some of these systems and not for others, so what could be searched and what could be confirmed are different extents. The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). “Narrowband” describes the assumed transmitter and not the resolving power of the archived channels. The targets were chosen by other programmes for other purposes, so every statistical statement here is about an archive-selected sample (§3.1).

## 2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern gener-

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ation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science (Morrison et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay & Tingay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA’s 10 receiver bands cover 35 to 950 GHz, and one of them has been searched for technosignatures. The lowest, 35–50 GHz, does hold observations of 4 stars of this sample, but all from one programme, all in a continuum mode whose 31.25 MHz channels are 2048 times coarser than the narrowest channel searched here, and all proprietary until 2027: the coverage exists and cannot carry a carrier search. The band above it holds none at all. For continuity with the centimetre literature, the Arecibo planetary radar radiated  $2.0 \times 10^{13}$  W at 2.38 GHz (Sheikh et al. 2025); an aperture of that diameter is not realisable in the millimetre band, so the illustrative benchmark of §6.3 is instead a dish of 12 m. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search of comparable bandwidth and the only comparable archival one; the earlier nearby-star searches of Steffes & DeBoer (1994) and Mauersberger et al. (1996) covered a single line. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a  $5\sigma$  flux threshold corresponds to  $\text{EIRP}_{\min} > 6.9 \times 10^{17}$  W. That target lies between 25 and 277 times further away than the stars of the carrier search here, so at a fixed received flux an EIRP ratio between the two surveys would be 3–5 orders of magnitude of  $d^2$  and almost nothing else. The quantity that does not depend on distance is the minimum detectable *received* flux,  $F_{\min} = S_{\min} \delta\nu$ , and it has to be compared at the channel width each experiment recorded:  $S_{\min}$  falls as  $\delta\nu^{-1/2}$  while  $F_{\min}$  rises as  $\delta\nu^{1/2}$ , so putting a coarse window on a fine channel arithmetically credits it with a depth it never reached. Mason et al. reach  $3.8 \times 10^{-23}$ – $5.8 \times 10^{-23}$  W m $^{-2}$  across their four fields, on a 30.52 kHz channel at a bare  $5\sigma$  flux threshold. On that same threshold convention and on native channels throughout, the windows searched here reach  $4.8 \times 10^{-24}$  W m $^{-2}$  at best — a factor 10 deeper, in a channel 15.3 kHz wide — and  $3.0 \times 10^{-22}$  W m $^{-2}$  at the median, a factor 6 shallower in a channel 488 kHz wide. Per channel this survey is therefore the shallower of the two except in its finest windows, and its advantage lies elsewhere: 60 stellar systems within 40 pc rather than distant stars in four calibrator fields, and a frequency

TABLE 1  
THE COMPARISON PROGRAMMES OF FIGURE 1, EACH ON ITS OWN DETECTION THRESHOLD AND ON THE COMMON  $5\sigma$  BASIS.  $\text{EIRP}_{\min}$  IS THE DEEPEST LIMIT EACH PAPER REPORTS, AND THE FACTOR IS  $5/(S/N)$ , THE MINIMUM DETECTABLE FLUX BEING LINEAR IN THE THRESHOLD IN EVERY ONE OF THEIR OWN EXPRESSIONS FOR IT.

| programme     | S/N | factor | $\Delta\nu_{\text{ch}}$ | $\text{EIRP}_{\min}$<br>(W) | at $5\sigma$<br>(W)   |
|---------------|-----|--------|-------------------------|-----------------------------|-----------------------|
| Enriquez + 17 | 25  | 0.20   | 2.8 Hz                  | $10^{13}$                   | $2 \times 10^{12}$    |
| Margot + 23   | 10  | 0.50   | 3.0 Hz                  | $1.35 \times 10^{13}$       | $6.75 \times 10^{12}$ |
| Price + 20    | 10  | 0.50   | 2.8 Hz                  | $2 \times 10^{12}$          | $10^{12}$             |
| Mason + 25    | 5   | 1.00   | 30.5 kHz                | $6.91 \times 10^{17}$       | $6.91 \times 10^{17}$ |

range reaching about nine times higher. Mason et al. also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §4.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. For a carrier narrower than every programme’s channel that limit is a total radiated power rather than a spectral power density, so the axis is commensurable. The detection thresholds behind the published limits are not: of the 4 programmes drawn, 3 quote their limits at a threshold other than  $5\sigma$ , and the minimum detectable flux is linear in that threshold in each of their own expressions for it. Table 1 gives each programme’s threshold, the factor that puts it on the common  $5\sigma$  basis and the channel width it used, and the figure is drawn after that rescaling, with each published value kept beside it. The survey’s own filled symbols are bare  $5\sigma$  thresholds on the same convention; the open symbols above them are  $\text{EIRP}_{90}$ , the power recovered nine times in ten through the single-epoch trigger stage, and every limit in this paper is quoted on that. Any EIRP advantage toward the nearest systems is a  $d^2$  effect and not an instrumental one. Panel (b) is the other axis, the native channel width of every window searched here beside the channel of each comparison programme. What distinguishes this survey is its frequency range and the proximity of its targets, not its depth in any one channel.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

Because of the very specific and detailed statistical nature of this study, we have extended the paper to include several very detailed Appendices that rigorously explain the methodology involved.

### 3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 280 h of telescope time and 1094 h

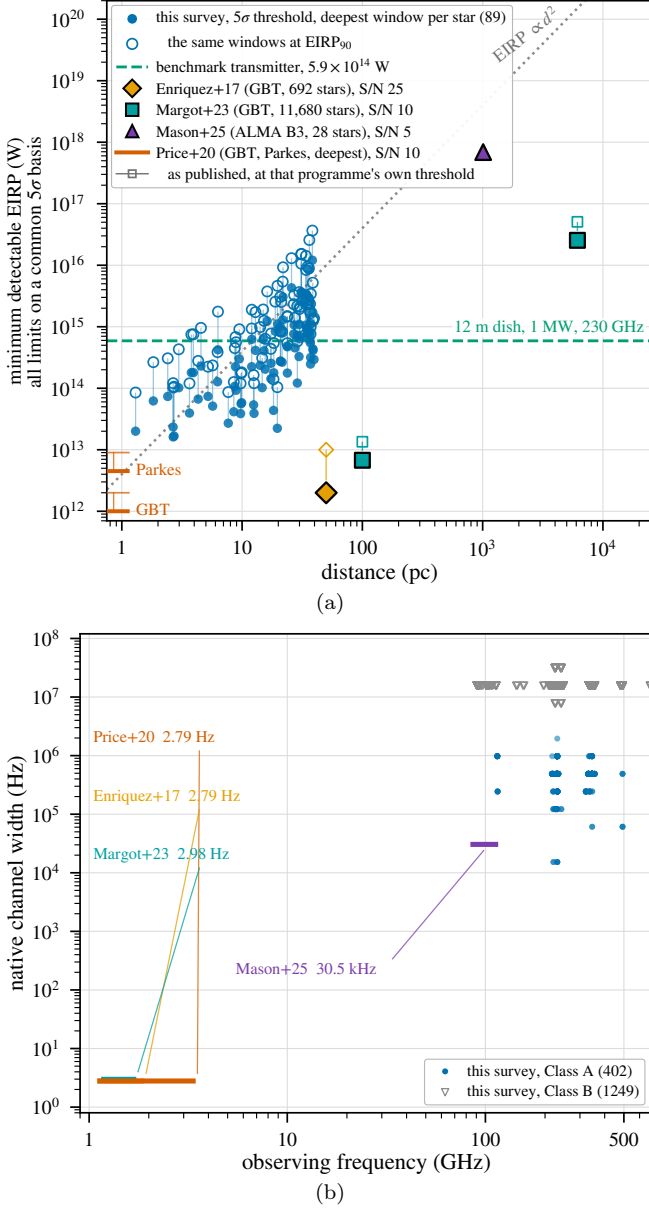


FIG. 1.— Where this survey sits among published radio technosignature searches. Every limit plotted is a total-power limit for a signal narrower than the native channel of the experiment that set it. Those channels differ by 7 orders of magnitude, from 2.79 Hz to 31.2 MHz, so the programmes are sensitive to different classes of spectral feature and a signal detectable by one of them need not be detectable by another at any power. (a) Minimum detectable EIRP against distance, all on a common  $5\sigma$  basis. Blue, this survey, the deepest window per star: *filled*, that window's bare  $5\sigma$  threshold, which carries no correction for the instrument's spectral response and so stands on the same footing as every other threshold in the panel; *open*, the same window at EIRP<sub>90</sub>, the single-epoch trigger completeness of  $\$5.1$ , which does carry that response, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020). Each of these is drawn at its published limit rescaled to  $5\sigma$  by the factor in Table 1, with the published value itself as the hollow marker joined above it. Grey dotted,  $\text{EIRP} \propto d^2$  at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of  $\$6.3$ . No symbol in the panel carries a confirmation criterion. (b) Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments at the widths of Table 1.

summed over windows. That set is the *primary census*, and every limit, every crossing and every expectation here is computed over it. Four further sets of blocks enter the paper without being part of it: a hold-out reserved before any block on its own side of the rule was searched, an archival extension searched afterwards, an out-of-sample control beyond 40 pc, and an external sample used only to calibrate the spatial statistic. Table A1 names all five in rows that sum, and sets beside them every count of windows, crossings, systems and bandwidth this paper quotes, each with the set it counts. Wherever a number here is computed over something other than the primary census, that table's name for it is used at the point of use.

*Star* means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments. 402 windows toward 65 stars in 60 systems are fine-channel — *Class A* throughout — and resolve a drifting carrier; every sensitivity and every exclusion quoted here refers to them unless stated otherwise. The remaining 1249 are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess, which cannot distinguish a drifting signal from a stationary one.

The inclusion criteria were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions,  $d = 1/\varpi$ , and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.5. It removes 8 windows, all of them  $\epsilon$  Eridani in Band 6, and no others; that band was then set aside as a whole, and Appendix A reports its limits in one place, including the 4 windows that place the star at the pointing centre and pass the floor. None of them enters a census statistic. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, inside even the stricter  $\sigma_\varpi/\varpi \leq 0.1$  of Table 2's reference census. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. One target has no catalogue name and appears throughout as ALMA J153702653–33192492, Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc.

Disc, planet and colour enter as descriptive metadata only, and no star was excluded by any criterion other than the three above. Three removals act on windows instead, and account for the whole difference between the 1721 rows the extraction wrote and the 1651 retained: 45 repeated rows of one window, the 12 withheld above, and 13 windows failing a quality cut that is physical rather than chosen. Since  $\sigma\sqrt{t_{\text{on}}}\Delta\nu_{\text{ch}}$  is set by system temperature and collecting area, a window more than 100 times below the sample median in it is not measuring the sky; those removed lie  $4.7 \times 10^3$ – $3.0 \times 10^4$  times below it and the nearest retained 24 times, so the threshold's value does not matter. No integration-time cut was applied; the shortest window retained carries 21 s on source.

### 3.1. The archival selection

This sample was not designed: it is whatever previous ALMA programmes happened to observe, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 2 measures the bias against a volume-limited reference census of which the searched stars are a subset, classifying both columns by one rule — BP–RP colour with the absolute G magnitude, which reaches 98 per cent of the census and 98 per cent of the searched sample — so every ratio below is like against like.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two under-representation beyond 30 pc. Colour comes second: A stars are over-represented by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. M dwarfs are depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.7 of their share of the reference census, and only 28 of them carry the fine-channel coverage that can resolve a drifting carrier at all. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The selection acts hardest on exactly the targets a designed search would pick: of the 15 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

### 3.2. Frequency coverage and archival scope

The Class A experiment covers  $\Delta\nu_A = 47.7$  GHz of unique sky frequency, and that coverage is sparse: 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band, so the endpoints are a range over which observations exist and never one that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. Fig. 2 shows the distribution; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the shortfall is accounted for. 177 further blocks lie inside the same query but were not processed when the census was frozen. 144 have since been calibrated and searched, 115 of them reported here as repeat epochs and as a sample the frozen pipeline had not seen; the other 29 enter no number anywhere, for the reasons Table A1 gives row by row. The largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all; no effort of ours would close it.

## 4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

### 4.1. The de-drifted search statistic

TABLE 2

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. BOTH COLUMNS ARE THE SAME GAIA DR3 SELECTION —  $\varpi > 0$ ,  $\sigma_{\varpi}/\varpi \leq 0.1$  AND  $1/\varpi \leq 40$  pc, 17 566 OBJECTS — OF WHICH THE 89 SEARCHED STARS ARE ROWS, SO THE TWO ARE SELECTED ALIKE BY CONSTRUCTION; AND BOTH ARE CLASSIFIED BY ONE RULE, IN WHICH BP–RP COLOUR SETS THE CLASS AND THE ABSOLUTE G MAGNITUDE SEPARATES THE DEGENERATES THAT COLOUR ALONE WOULD PLACE AMONG THE A STARS. THE RULE REACHES 17 278 CENSUS OBJECTS AND 87 OF THE 89 SEARCHED STARS, THE REST CARRYING NO GAIA BP OR RP PHOTOMETRY. “RATIO” IS THE SEARCHED COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION;  $> 1$  IS OVER-REPRESENTATION. THE STRONGEST SELECTION EFFECT HERE IS DISTANCE, NOT COLOUR.

| Property                            | Census         | Searched   | Ratio |
|-------------------------------------|----------------|------------|-------|
| Distance                            |                |            |       |
| $d \leq 5$ pc                       | 48 (0.3%)      | 12 (13.5%) | 49.3  |
| 5–10 pc                             | 240 (1.4%)     | 13 (14.6%) | 10.7  |
| 10–20 pc                            | 2069 (11.8%)   | 18 (20.2%) | 1.7   |
| 20–30 pc                            | 5391 (30.7%)   | 19 (21.3%) | 0.7   |
| 30–40 pc                            | 9818 (55.9%)   | 27 (30.3%) | 0.5   |
| Colour class <sup>a</sup>           |                |            |       |
| A                                   | 61 (0.4%)      | 5 (6.3%)   | 17.4  |
| F                                   | 473 (2.8%)     | 8 (10.1%)  | 3.6   |
| G                                   | 978 (5.8%)     | 15 (19.0%) | 3.3   |
| K                                   | 2221 (13.2%)   | 10 (12.7%) | 1.0   |
| M                                   | 13 082 (77.8%) | 41 (51.9%) | 0.7   |
| Other axes                          |                |            |       |
| Confirmed planet hosts <sup>b</sup> | 26 (15.5%)     | 23 (25.8%) | 1.7   |
| White dwarfs <sup>a</sup>           | 463 (2.7%)     | 8 (9.0%)   | 3.4   |
| Searched stars / systems            | —              | 89 / 82    | —     |

<sup>a</sup>Colour bins, not MK types. The boundaries are the colours at which the census’s own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP–RP  $< 0.36$  for A,  $0.36$ – $0.72$  for F,  $0.72$ – $0.95$  for G,  $0.95$ – $1.77$  for K and  $> 1.77$  for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 79 non-degenerate searched stars and the 16 815 non-degenerate census objects.

<sup>b</sup>Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 23 searched hosts carry 50 confirmed planets, 21 of the 23 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

Each window’s dynamic spectrum is stacked along every trial drift rate  $\dot{\nu}$ . The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where  $I(\mathbf{x}, t, \nu)$  is the spectrum at sky position  $\mathbf{x}$  in integration  $t$  and channel  $\nu$ , and  $\sigma(t, \nu)$  is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so  $T$  is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. The statistic

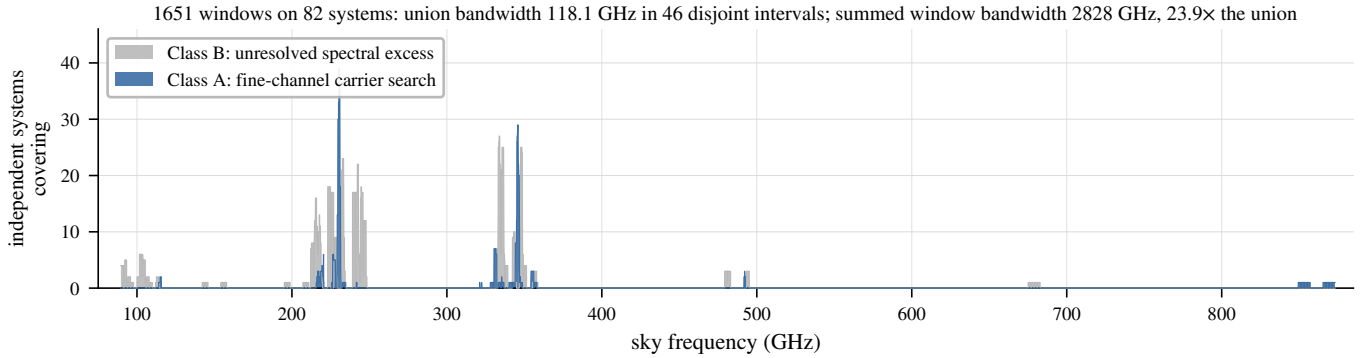


FIG. 2.— Where this survey searched. *Lower panel:* each searched window is a horizontal segment on its host system’s row, rows ordered by distance; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel:* the number of systems covering each gigahertz of sky frequency. The summed window bandwidth is 23.9× the union, so the union measures frequency coverage and the summed bandwidth search effort.

is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable.  $T_*$  denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean  $I = (XX + YY)/2$  of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star’s own frame and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity.

Two power quantities carry the main text, both defined in Table 3: the trigger level  $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$  at distance  $d$ , for a carrier that makes  $T_*$  reach  $T_{\text{trig}} = 5$  in one native channel of width  $\Delta\nu$ , and the sensitivity  $\text{EIRP}_{90}$ . The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. Three campaigns did that, each with its own denominator. The completeness itself is measured on 49 of the 402 fine-channel windows, injected directly, and on 5936 carriers over 35 windows of the coarse class. What the

spatial rank costs is measured separately, on 8860 carriers over 52 of 56 attempted block-window configurations. The primary-beam correction is tested on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the  $\text{EIRP}_{90}$  of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 4. Two comparisons bear on the amplitude scale and they are not the same kind of check. An independent reduction of the same AU Mic visibilities is reproduced to  $0.994 \pm 0.014$ , which tests the extraction and not ALMA’s flux transfer, an error in which would appear in both reductions alike; the external comparison is with the published quiescent flux density of  $\epsilon$  Eridani, recovered to  $1.10 \pm 0.31$  (Burton et al. 2022) and so a 28 per cent measurement. The absolute scale is therefore carried at ALMA’s own band-dependent accuracy specification, and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, accordingly carry  $\pm 21$  per cent where the sample as a whole carries  $\pm 11$ .

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot{\nu}}{\nu} = \frac{a_{\text{los}}}{c} \quad \text{and so} \quad a_{\text{los}} = \frac{c \dot{\nu}}{\nu}, \quad (2)$$

so a ceiling expressed as  $\dot{\nu}/\nu$  is the same physical limit in every band. The drift and acceleration ceilings adopted for each window were computed from Equation 2 window by window, and reproduce it to  $1 \times 10^{-4}$  over all 402 Class A windows. The default ceiling is 12.0–13.3  $\text{Hz s}^{-1} \text{GHz}^{-1}$ , which is 12.0–13.3 nHz in the fractional units the literature uses, and is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more; at unit margin the search would have been blind to two of the windows that produced a crossing. It is 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend, their figure being set by the rotation of small, fast solar-system bodies rather than by a planet’s own orbital acceleration, and the restriction here is deliberate. The grid is uniform in  $\dot{\nu}$ , and fine enough that a carrier midway between two trials loses no more than 0.26 of a

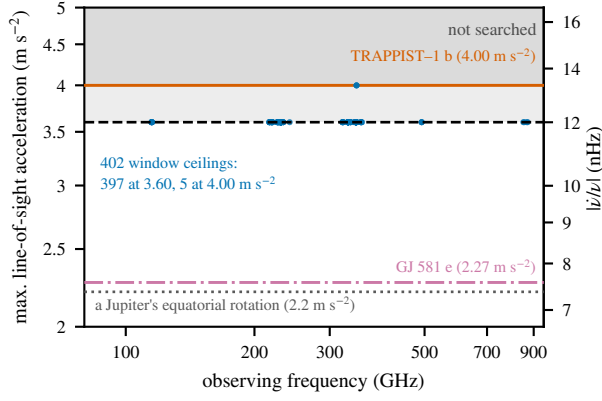


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration by Equation 2, against observing frequency; the shaded region above is unsearched. The right-hand axis is the same quantity as a fractional drift rate,  $|\dot{\nu}/\nu| = a/c$ , which is the unit the literature states drift reach in. The boundary drawn is the default ceiling,  $3.60 \text{ m s}^{-2}$ , equivalently  $12.0 \text{ nHz}$ , which is also the median; it runs to  $4.00 \text{ m s}^{-2}$  on the 5 TRAPPIST-1 windows. Horizontal lines are reference accelerations: the two most demanding known planets of any target, TRAPPIST-1 b ( $4.00 \text{ m s}^{-2}$ ) and GJ 581 e ( $2.27 \text{ m s}^{-2}$ ), and the equatorial rotation of a Jupiter,  $2.2 \text{ m s}^{-2}$  or  $7.4 \text{ nHz}$ . Earth’s own rotation and orbit,  $0.034$  and  $0.0059 \text{ m s}^{-2}$ , lie below the foot of the axis. **Transmitters whose combined orbital and rotational acceleration exceeds the ceiling are not constrained by this search, and the ceiling is a budget they share:** a transmitter on the equator of a rotating giant spends 61 per cent of it on the planet’s spin alone. Nor are unbound transmitters, or any platform accelerating for its own reasons.

native channel over the longest track. Three limits follow: a deliberately chirped carrier is not distinguished from an accelerating platform; a transmitter need not be bound to a known planet, so drifts faster than the ceiling are unconstrained; and orbital and rotational acceleration add, so a transmitter on a rapidly rotating body reaches the ceiling on its spin alone. For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened grid used on that star’s own windows (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 3). Channelisation in the sample is bimodal with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track,  $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta \nu_{\text{ch}} \gtrsim 1$ . Each window’s own  $\eta_{\text{drift}}$ , trial-drift count and acceleration ceiling follow from its channel width, its track length and Equation 2, so the sample may be reclassified on that criterion instead; the two do not partition the survey identically. The two classes are physically different experiments, results are given per class, and they are combined only where that is appropriate.

#### 4.2. From a crossing to a confirmed signal

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 4). Each is set out below as the quantity it measures, the number that

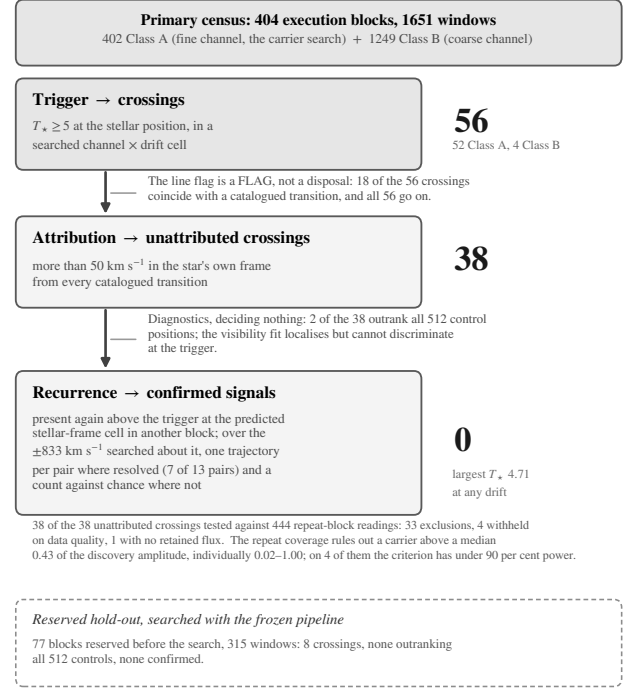


FIG. 4.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and the quality-failing block that no expectation is computed over, so the figure, Table D1 and the text cannot disagree. The counts drawn are the primary crossings, one to a window; the 21 secondary crossings of §5.3 enter the same three levels and three reach the second, none the third.

quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger → a crossing.* The quantity is  $T_*$ , the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is  $T_* \geq T_{\text{trig}} = 5$  in at least one channel  $\times$  drift cell; and both the statistic and the level were fixed before the archival search began.  **$T_{\text{trig}}$  is a trigger threshold and not a detection significance:**  $T$  is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between  $112$  and  $6.87 \times 10^6$  of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches the level is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line. A crossing establishes only that it goes on to the next two tests.
- (ii) *Line attribution → an unattributed crossing.* The quantity is the crossing’s velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star’s own rest frame; the criterion is

$\pm 50 \text{ km s}^{-1}$ . The statistic and the trigger were fixed before the archival search began; the line catalogue is a single database query made afterwards, on 2026 October 6, so the dispositions rest on the physical evidence set out beside each one and not on the order in which the list was assembled. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window's price is that the survey constrains nothing inside it (Appendix C).

- (iii) *Independent recurrence  $\rightarrow$  a confirmed signal.* The quantity is  $T_*$  at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

*Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything.* The first ranks  $T_*$  against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block's own synthesised and primary beams so that no control sits on the star; its lowest value,  $1/(512 + 1)$ , means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as  $\beta$  Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing's emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for

the rank would give. Throughout,  $p$  denotes a genuine significance level, and no probability computed against the rank's own null is called one.

*One execution block is not a null, and a uniform rule says so.* The comparison with the control positions needs each block's ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002\_Xc7fa6f\_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. The Class A results are identical with it and without it, the same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey's entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window's own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by  $\beta$  Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

## 5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the primary census. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

### 5.1. Sensitivity: the power at which a carrier would have been found

*The sensitivity of this survey is a recovery completeness of one epoch, and not a threshold.* Artificial carriers were injected at a random sub-channel phase and a random drift from each window's own grid and recovered through the unchanged pipeline; the power at which nine in ten come back is  $\text{EIRP}_{90}$ , the only sensitivity quoted here, measured through the single-epoch trigger and the localisation at the stellar position inside one execution block. **It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed**, because confirmation needs a second and separated epoch, which the archive supplies for some systems

TABLE 3

THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER. THE DISTINCTION THE FIRST THREE ROWS DRAW IS THE ONE THE LIMITS OF §5.1 TURN ON: A CROSSING IS A SINGLE-EPOCH EVENT, AND A CONFIRMED SIGNAL NEEDS A SECOND AND SEPARATED EPOCH THAT NO SENSITIVITY SUPPLIES.

| Term                  | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crossing              | A channel $\times$ drift cell at the stellar position reaching $T_* \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its <i>primary</i> crossing and any other a <i>secondary</i> one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Unattributed crossing | A crossing lying more than $50 \text{ km s}^{-1}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Confirmed signal      | An <i>unattributed</i> crossing recovered at the predicted stellar-frame frequency and drift in an independently separated block. The criterion is applied to every crossing, and a crossing coincident with a catalogued transition that satisfies it is a recurring molecular line: 13 of them are, which is this stage's positive control (§5.6). The reading is taken at one channel at one drift, so it holds only for a carrier whose stellar-frame frequency moved by between $\pm 0.08$ and $\pm 0.63 \text{ km s}^{-1}$ at most between the epochs compared; over the $\pm 833 \text{ km s}^{-1}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory where the drift grid resolves the interval between them (7 of 13 pairs, Table 5). It can be applied only where the archive observed the same frequency again, so for the 23 Class A systems where it did not, no sensitivity makes a confirmed signal possible. |
| $P_{\text{trig}}$     | The power of a carrier that makes $T_*$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| EIRP <sub>90</sub>    | The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 353 an estimate from the class-average calibration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Class A               | The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Class B               | The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| The sample            | Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

and not others whatever their depth (§5.6.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 5).

*Measured that way the completeness is one number for the class.* Nine carriers in ten are recovered at  $3.03 P_{\text{trig}}$ ; the sampling error of that pooled value is  $\pm 2$  per cent, which is a statement about the class average and about no individual window. It is a total power set against a nominal threshold:  $P_{\text{trig}}$  is the bare  $5\sigma$  of the window itself, and the loss a channel-confined carrier suffers to the correlator's spectral response is carried by the injected tones, so it sits inside the measured factor —  $\times 2.29$  of it — and is not applied to the result as well (Appendix A). Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring: the 5 windows whose ring carries resolved emission come back at  $3.24 P_{\text{trig}}$  against 2.94 for the rest (Fig. 5). Class B carries its own factor  $4.17 P_{\text{trig}}$ , measured the same way from 5936 carriers injected into 35 coarse windows of 14 execution blocks; it is what sets the limits for Barnard's Star, Wolf 359 and Proxima.

*The resulting limits.* The median Class A window reaches  $2.9 \times 10^{15} \text{ W}$  of equivalent isotropic radiated power and the median system, on its best window,  $1.6 \times 10^{15} \text{ W}$ , with a spread across systems of  $8.7 \times 10^{13} - 3.7 \times 10^{16} \text{ W}$  (Fig. 6a). The round  $10^{15} \text{ W}$  level is reached for 24 of them. The calibration of the factor is good to  $\times 0.89 - \times 1.11$  (Table 4); that is the interval on the median and on the counts, and not on any one window's limit. One bias is not inside it, runs one way only, and is *not* included in the values quoted here: an injected carrier is added to already calibrated visibilities and so suffers no atmospheric decorrelation, whereas a real carrier does. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §10.4.9), through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median limit optimistic by 1 per cent and the median window above 400 GHz by 8 per cent, so the limits at the top of the frequency range are the least secure quoted here. **The 125 blocks whose delivery exposes no such record are not a random subset of the 404:** their longest baseline has a median of 279 m against 46 for the 279 that do, and it is the long baselines that lose coherence. Matching each of their windows to the measured blocks within a factor of two in baseline length puts the correction at 1.9 per cent in the median and 21 at the ninetieth percentile, so their limits are conditional on a calibration performance the archive does not report.

*Two further losses act window by window, and both are divided out of the limits above rather than quoted beside them* (Appendix A). The search's phase centre omits the annual parallax, which displaces the star by at most 0.67 of a synthesised beam and makes the median Class A limit shallower by 1.5 per cent, while Wolf 28 in A002\_X9f2ff8\_X25b0 has no astrometric keys and is left uncorrected. That correction is concentrated rather than general, which is why the median limit moves far more than the median window's own factor (Table 4): 216 of the 402 windows need none at all, while 14 lose more than a tenth and lie around the middle of the dis-

tribution of limits. The second loss the injections do not reproduce at all, each carrier being laid down at one frequency per integration while a real one sweeps within it. **It costs nothing in the median window and up to  $\times 1.53$  in the 13 finest-channel ones, which are also the deepest:** the best Class A window moves from  $6.8 \times 10^{13}$  to  $8.7 \times 10^{13}$  W, and HIP 10679 drops out of the 25 systems that would otherwise reach  $10^{15}$  W.

*Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above.* The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance and without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild, so band and array were fixed by the draw and the conditions that vary within a block were compared with the population afterwards. **That comparison covers 6 properties** — channel width, integration time, noise, controlling brightness, the channels a carrier at the drift ceiling crosses in one track, and the intra-integration retention — and on every one the injected windows span the central eight-tenths of the population, five agreeing in the median to 6.3 per cent. The sixth, the noise level, does not: the injected windows are  $\times 1.7$  noisier, which follows from drawing whole blocks, and it is also the property the measurement is built to be insensitive to, each ladder being in units of its own window's threshold. That is measured rather than assumed: splitting the injected set at the median of each property in turn moves the ninety per cent point by at most  $0.27 P_{\text{trig}}$ . The draw controlled the array, but the achieved mix still departs from the population by 22 per cent of the sample; re-weighting the pooled curve by the survey's own composition gives  $2.98 P_{\text{trig}}$  against the adopted 3.03.

**A window that was not injected therefore carries an estimated EIRP<sub>90</sub>**, and across the injected windows the ninety per cent point ranges from  $\times 0.80$  to  $\times 1.58$  of the adopted factor: that dispersion, not the calibration interval above, is what one uninjected window's own limit is uncertain by. Because a system's limit is its best window, a minimum over several such estimates could favour windows whose own factor falls below the class average; drawing each window's factor from the 49 measured ones instead, over 4000 trials, moves the per-system median deeper by 2.2 per cent, holds the count at  $10^{15}$  W to 23–24 in nine trials in ten and spreads one system's own limit by  $-17$  to  $+9$  per cent, so the estimate is mildly conservative. **But the best window was itself injected into for only 6 of the 60 systems:** for the other 54 the headline limit is an estimate.

*A carrier that is not always radiating is penalised in proportion.* The de-drifted stack is coherent across a whole execution block, so over 3888 trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent: a transmitter radiating a tenth of the time needs  $10\times$  the power quoted above. Those trials carry no drift, so the joint drifting and intermittent case is not measured here.

TABLE 4  
WHAT SETS EIRP<sub>90</sub>, AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; ITS FLUX TERM IS BAND-DEPENDENT, SO THE SUBMILLIMETRE WINDOWS CARRY THE LARGER COMBINED FIGURE. THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW'S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR, AND THE DECORRELATION TERM IS MEASURED IN 279 OF THE 404 BLOCKS AND NOT IMPUTED TO THE REST.

|                                                                                     | value                             | what it does to the limit                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class-average completeness                                                          | $\times 3.03$                     | nine carriers in ten recovered at this multiple of a window's own trigger power. <i>Measured</i> in the 49 Class A windows injected directly and <i>estimated</i> , at this factor, in the other 353                                                                                                                        |
| <i>Uncertainty on that calibration (per cent), combined in quadrature</i>           |                                   |                                                                                                                                                                                                                                                                                                                             |
| The factor itself                                                                   | $\pm 2$                           | sampling error of the pooled value over the 49 injected windows                                                                                                                                                                                                                                                             |
| Absolute flux scale                                                                 | $\pm 10.2$                        | the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): $\pm 5$ in Bands 1–5; $\pm 10$ in Bands 6 and 7; $\pm 20$ in Bands 8–10, weighted over this sample's bands |
| Distance and visibility scale                                                       | $\pm 4.5$                         | the Gaia parallaxes, which enter as $d^2$ , and the block-to-block visibility scale                                                                                                                                                                                                                                         |
| Combined                                                                            | $-11/+11$                         | $\times 0.89\text{--}\times 1.11$ on the factor, and so on the median limit and on the system counts; $\pm 21$ for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz                                                                                                                                     |
| <i>Spread between windows. An observed dispersion, not an error bar.</i>            |                                   |                                                                                                                                                                                                                                                                                                                             |
| Window to window                                                                    | $\times 0.80\text{--}\times 1.58$ | the 49 injected windows resolve $2.41\text{--}4.80 P_{\text{trig}}$ individually. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 54 of the 60 systems                                                                                                         |
| <i>Applied. Folded into every limit quoted, and so not combined with the above.</i> |                                   |                                                                                                                                                                                                                                                                                                                             |
| Intra-integration smearing                                                          | $\times 1.00\text{--}\times 1.53$ | a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\text{smear}}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1046 windows) and from the longest delivered anywhere where it is not (605)           |
| Annual parallax                                                                     | $\times 1.00\text{--}\times 2.06$ | the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent                                                                                                                                                                               |
| <i>Signed terms. A bias is not an error bar and is not added to one.</i>            |                                   |                                                                                                                                                                                                                                                                                                                             |
| Channel response                                                                    | $\times 2.29$                     | the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of $\times 1.33$ over the nominal trigger                                                                                                          |
| its residual                                                                        | $+8$                              | the injected profile falls short of the exact response: <i>conservative</i>                                                                                                                                                                                                                                                 |
| Decorrelation                                                                       | $+0/+1$                           | measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does: $+8$ above 400 GHz, and more than $+20$ in 5 per cent of windows                                                                                        |

## 5.2. The nearest stars: Barnard's Star and Wolf 359

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B*

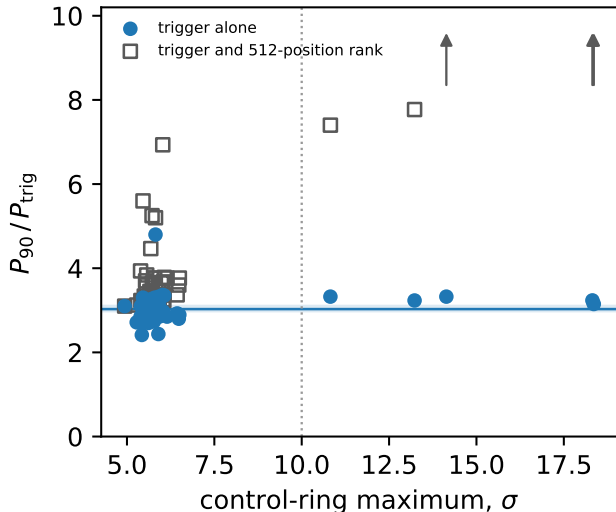


FIG. 5.— Why one completeness factor serves the whole class. Each of the 49 of the 402 Class A windows injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the  $10\sigma$  ring brightness above which a ring carries resolved emission rather than noise.

*experiment and not from the carrier search:* the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the  $\text{EIRP}_{90}$  convention of §5.1 Barnard’s Star reaches  $2.7 \times 10^{14}$ – $4.0 \times 10^{14}$  W over 12 windows and Wolf 359  $3.1 \times 10^{14}$ – $4.5 \times 10^{14}$  W over 12.

### 5.3. Threshold crossings and how each was disposed of

*The result is a chain of three tests, and it is read in one direction.* Of the windows the search covers — the 1651 of the primary census and the archival-tail blocks beside them — 56 hold a cell reaching  $T \geq 5$  at the stellar position. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population that characterises the search is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask,  $\pm 50 \text{ km s}^{-1}$  wide in each star’s own rest frame, places 21 of the 52 within that distance of a catalogued transition; three of those coincidences are with a species no debris disc has been shown to carry and are reported rather than acted on (Appendix C), leaving 18 attributed and 34 unattributed. The attributions are kinematic and not an accident of bandwidth: 11 of the 51 crossings with a measured peak frequency fall within  $\pm 50 \text{ km s}^{-1}$  of a searched transition, against 4.7 expected were each peak free to land anywhere on its own window’s grid (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 4 counts what survives each step and Tables D1 and D2 list every crossing.

*Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too.* The de-drifted statistic is retained in every channel, so be-

hind a window’s tabulated crossing — its largest cell — weaker above-trigger cells can be read as well, neighbouring channels counting as one. Re-read that way, the 399 of the 402 Class A windows that retain such a profile — the other 3 retain none — hold 49 maxima at or above the trigger and twenty-one weaker cells besides, in 9 windows and as many as 8 in one. 18 are the resolved carbon monoxide of a disc spread over channels; the three unattributed ones, toward HD 14055, 61 Vir and  $\beta$  Pic, each have 3–16 covering blocks and none reaches the trigger in any of them, the deepest repeat being  $T_{\star} = 4.4$ . Applied to the window maxima the rule returns 50 of the 56 tabulated crossings to 0.002 of a channel; the 6 it misses are the 4 of the failing block, whose repaired profile was not retained, and crossings whose own window retains no usable profile. Cells are counted in the controls too, and outside the mask the star holds 1.05–1.15 times as many as a control position at every level from  $T = 3$  to the trigger, with no step where a carrier search would put one; the expectation that follows is §5.3.1’s.

#### 5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel  $\times$  drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the 402 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window’s own 512 control positions measure how often a position in that field reaches the trigger, and the window’s own channel grid measures how often such a position would fall inside an attribution window, which takes 20 per cent of the total. The control positions differ from the star in one respect that bears on this: they carry no continuum, so a calibration error that varied with frequency would raise the star and not them. Measured on the continuum these stars are found to have, the windows where such an error could reach the trigger hold no crossing and bound it below 6.6 per cent (Appendix B). Chance predicts 28.9 unattributed Class A crossings, with 12–48 allowed by resampling the execution blocks the windows come from, and 32 are found. Counting cells rather than windows on both sides moves the two together: the same 3192 control positions predict 28.1 cells, 2.2 of them beside a window’s largest, against 34 and 3 found and 13–48 allowed, because a control position holds a second above-trigger cell about as often as a star does (§5.3). A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

The expectation is a sum of rates measured at control positions, and the reserved blocks measure how far a star departs from them: carrying that difference through moves it to 30.8, and the count agrees with chance either way. The width of the range allowed is what the comparison can say and no more — an added population of as many as 19 crossings would sit inside it unremarked. What tests each crossing individually is the recurrence test of §5.6.

#### 5.3.2. One execution block accounts for four of the crossings

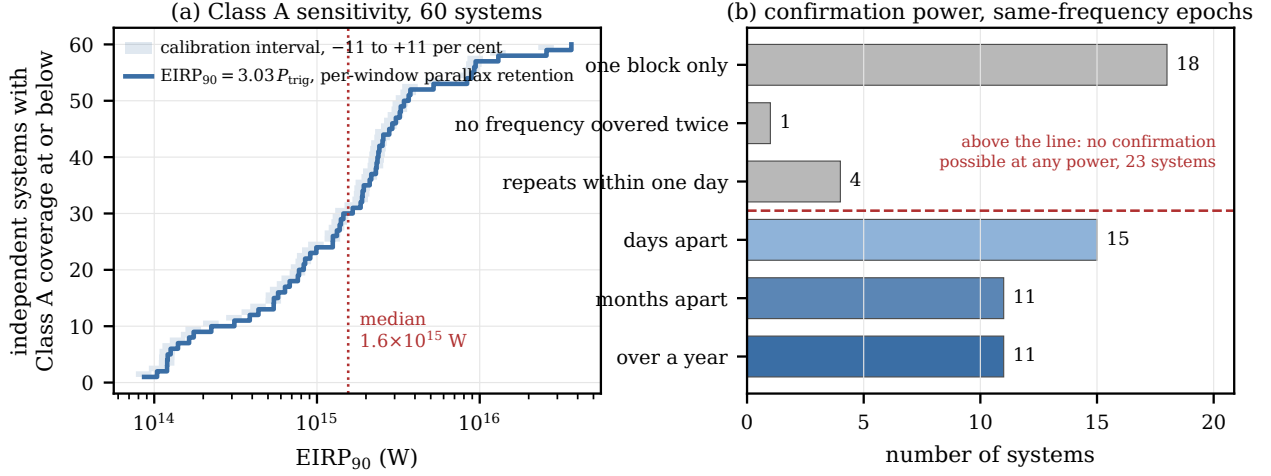


FIG. 6.— **The two completenesses of the carrier search, over the same 60 systems.** (a) Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given EIRP<sub>90</sub> (Table 3), evaluated as  $3.03 P_{\text{trig}}$  with each window’s own annual-parallax retention applied; the shaded band is the calibration interval of Table 4. (b) Confirmation completeness, which no amount of power buys: systems grouped by the longest separation between two execution blocks covering a common frequency, more than 1 day counting as independent. **Left of the dashed line, 23 systems have no such pair, and for them confirmation was not possible at any sensitivity in panel (a): what panel (a) gives them is a crossing limit, not a detection limit.**

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all in one ACA execution block, A002\_Xc7fa6f\_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed  $T = 5$  against a survey median control maximum of 2.75. Its single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, while the 24 sibling executions of the same field, array and setup reach only  $T_* = 4.26$  and produce no crossing. *What is inflated is the field and not the star:* the strongest of the four reaches  $T_* = 73.4$  where its own controls peak at 74.7, and 1–421 of the 512 stand above the star. The criterion fails this block and no other (Appendix D.2), so its 4 crossings are reported apart from the 52; Appendix E carries what the block might instead be.

#### 5.3.3. Stellar flares are excluded by a broadband prediction

The sample contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a  $5\sqrt{N}\sigma$  continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4,717 $\sigma$ , against an observed continuum statistic of median -0.29 with  $|Z| < 3$  in 45 of them. One block settles it: AU Mic A002\_Xa42f75\_X319 holds both a crossing ( $T_* = 5.27$ ) and the millimetre flare of MacGregor et al. (2020), and that flare contributes 0.17 $\sigma$  to the single-channel statistic — raising the continuum at  $Z = 10.3$  while the crossing’s own de-drifted track reaches  $Z_{\text{max}} = 2.1$ .

#### 5.4. $\beta$ Pictoris: an astrophysical positive control, and the obstacle it identifies

*A search that finds nothing must show that it could have found something, and the survey’s strongest features show exactly that.* The frozen pipeline recovers the  $\beta$  Pictoris circumstellar carbon monoxide blind, without being

told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at  $T_*$  up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block’s own epoch, then to the star’s rest frame — all 10 crossings lie within 0.5 km s<sup>-1</sup> of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block’s own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged  $\beta$  Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning  $T_* = 8.95$  in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

*One of those blocks is a counterexample, and it is the more useful half of the control.* In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic’s reach as well as demonstrating it (Fig. B1).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches  $T_* = 26.83$ , but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward

the source and flag as possible foreground cloud emission. The block's  $^{13}\text{CO}(2\rightarrow 1)$  window is bright at the ring and faint at the star. What identifies the extended emission is therefore the *level* of the control ring and not the star's rank against it — the star here outranks every control — which is why the window-level flag of Appendix D.2 is defined on the ring's median.

*Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity.* The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 18 of the 56 crossings land on a catalogued transition. The mask that removes them costs 6.4 per cent of the Class A union bandwidth over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

### 5.5. The two crossings that lead their own control fields

Frequencies in this section and the next are in the star's own rest frame, which is the frame the recurrence test registers in; the delivered ones are in Table 6 and in the ledger.

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.3232 GHz and CP–72 2713 at 344.2965 GHz, both in Band 7. They are drawn in Fig. 7 and tabulated in Table 6, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4). CP–72 2713's crossing lies  $-12\text{ km s}^{-1}$  from  $\text{SO}(8_8-7_7)$ , a coincidence reported as a note rather than as a disposition (Appendix C). Both events go on to the recurrence test. Each was also re-fitted in the visibility domain, as 51 of the 56 crossings were, by phase rotating to the star and following the recovered drift; 41 of those fits are consistent with a point source at the star, and injected sources in the bin these two occupy return  $\text{Re}/\sigma = 6.58 \pm 1.97$ , so the fit excludes none. Its value is that a pedestal fed through a sidelobe from elsewhere in the field, which a dirty map at the star cannot tell from a source at the phase centre, is not a point source there. Of the 512 control positions the rank screen uses, 8 retain a dynamic spectrum and 12 are reachable in the visibilities.

Neither is present when the same stellar-frame frequency is observed again. The reading that assumes least is the nearest covering block alone, taken 1.97 and 2.05 hours from each discovery: a carrier persisting at the discovery amplitude would have given  $T_{\text{pers}} = 5.28$  there against  $-0.01$  measured for 61 Vir and 6.2 against  $-0.57$  for CP–72 2713 — whose nearest block is also its deepest — ruling out anything above 0.95 and 0.71 of that amplitude. Over intervals of hours a transmitter orbiting at 1 au stays inside the  $\pm 0.211$  and  $\pm 0.213\text{ km s}^{-1}$  half-channel those readings cover, so both survive an orbit of that period. Combining all 3 and 5 covering blocks deepens the limits to 0.56 and 0.66 of that amplitude, but requires the carrier to hold one channel across baselines of 3 and 2 days, over which the same orbit would move it  $1.53$  and  $1.08\text{ km s}^{-1}$ : those readings constrain only a carrier stable to half a channel over the whole baseline. At the faintest amplitude consistent with each crossing having been found at all, the criterion would have recov-

ered a persisting carrier with probability 1.00 and 0.66: for CP–72 2713 the silence is not an exclusion, and is not read as one. Appendix B follows CP–72 2713's crossing through every step of the chain.

### 5.6. Recurrence, applied to every unattributed crossing

A carrier that stays inside the repeat window and radiates through both observations must be there when the same frequency is observed again. That is the confirmation criterion, applied in the star's own frame: the predicted channel follows from the Earth's motion at the repeat epoch and the star's systemic velocity, the discovery drift is carried to it, and the statistic is read there, maximised over nothing. The frame matters, because for  $\eta$  Crv it moves the predicted channel by  $-80.2$  channels. A *confirmed signal* is an *unattributed* crossing that satisfies the criterion (Table 3); a line-coincident crossing that satisfies it is a recurring molecular line, and is reported as one.

None of the unattributed crossings recurred, while 13 crossings attributed to circumstellar or foreground carbon monoxide do recur, as expected. Those 13 are the positive control of this stage and they test it on both of the ways real emission arises here: 10 are the resolved circumstellar disc of  $\beta$  Pic and 3 are foreground gas toward HD 285968, whose sightline carries catalogued cloud emission at the crossing's own velocity in the local standard of rest (Appendix C). No exclusion is formed for them, because what one would exclude has just been observed. Close to the trigger the control is thinner than that count suggests: every circumstellar recovery has  $T_{\star} \geq 8.0$ , and the 2 attributed crossings that are not recovered are the two faintest tested, both at  $T_{\star} = 5.41$ , so there the stage is controlled by HD 285968 alone — by foreground gas rather than by a point source. All 38 unattributed crossings were tested, over 444 readings in 151 execution blocks, and at the predicted cell the largest statistic over all of them is  $+2.00$  at the carried drift and  $4.71$  over every drift trial, against a trigger of 5.

Every exclusion is conditional on a requirement of frequency stability, and it is a narrow one. The reading is taken at one cell at one drift, so it constrains only a carrier whose stellar-frame frequency moved by less than half that cell between the epochs compared:  $488\text{ kHz}$ , which is  $\pm 0.08$  to  $\pm 0.63\text{ km s}^{-1}$  across the 33 crossings that carry an exclusion, median  $\pm 0.21$ . A transmitter in a circular orbit at 1 au accelerates along the line of sight by up to  $5.9\text{ mm s}^{-2}$ , which keeps it inside that interval for 14 of the 33 over the interval to the nearest covering block and for 1 of them over the longest baseline; at the  $3.6\text{ ms}^{-2}$  the drift grid reaches, none of the exclusions survive. The search accommodates planetary acceleration and this test does not, so the two bear on different classes of transmitter: non-recurrence excludes a carrier that holds its frequency within those limits and radiates through both observations, and says nothing at all about one whose orbital motion carries it out of the cell. That is why the whole window is searched at every drift of the grid at each repeat epoch, which spans that excursion for 30 of the 32 crossings another block covers.

The amplitude each exclusion reaches is the fraction of the discovery amplitude still excluded at the trigger,  $(T_{\text{rep}} + 5)/T_{\text{pers}}$ . Of the 38 crossings tested, 33 carry a quoted exclusion; 4 take their discovery amplitude from

the block that fails the quality criterion of §5.3.2 and 1 retains no discovery spectrum, so none can be formed for those. Over the 33 the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude and the deepest at 0.017 of it, but the individual limits run to 1.00, and for 3 of them —  $\beta$  Pic at 241.77 and 241.86 GHz and HD 110411 at 219.44 GHz — nothing below 0.84 is ruled out. Combining every covering block buys most of that, and requires the carrier to return to the same cell at the start of each: the nearest block alone rules out a median 0.98 of the discovery amplitude, and 15 of the 33 rule out nothing. The chance that a carrier would have satisfied the criterion is evaluated at the trigger amplitude  $5S/T_*$  rather than at the discovery amplitude, which is selected as the largest of many cells at a statistic of about five and is biased high (Appendix B): it exceeds 90 per cent for 26 of the 33 and for the remaining 7 it is  $\beta$  Pic at 241.86 GHz (0.38), ALMA J153702653–33192492 at 347.19 GHz (0.50),  $\beta$  Pic at 241.77 GHz (0.61), CP–72 2713 at 344.30 GHz (0.66), CD–57 1054 at 346.53 GHz (0.82), AU Mic at 230.67 GHz (0.85) and HD 110411 at 219.44 GHz (0.88). Those 7 are untested rather than excluded. These are conditional detection probabilities under the persistent-carrier model adopted here, not probabilities that a candidate was noise: a crossing that does not recur has not been shown to be anything else.

The 28 repeat windows that do hold something above the trigger are themselves crossings, so a carrier present again would appear in its repeat block as one; their number agrees with the 28 chance predicts in trials of that size, but each pair must still be asked whether one transmitter could have produced both members. Table 5 asks it of all 13 pairs, across 5 stars, whose two crossings lie within  $833 \text{ km s}^{-1}$  of each other in one star’s frame. Both fitted drifts are carried into that frame first, by removing the observatory’s own line-of-sight acceleration: at most  $40 \text{ Hz s}^{-1}$  here, but diurnal, so extrapolating it across days would add an excursion that is not there.

Two criteria follow, and they partition the pairs differently. Where the drift grid resolves the interval — 7 of the 13 pairs — the elapsed time and the two drifts fix the offset a carrier at constant line-of-sight acceleration would show, and all 7 fail that comparison, by 3.6 standard deviations in the mildest case. The weaker demand is any trajectory whose drift stays inside the searched grid and whose curvature is bounded by the same acceleration ceiling, and it is the one each verdict rests on: it excludes 3 pairs — 1 needing a mean drift the search never covered and 2 needing an acceleration reversal 1.3 to 2.4 times faster than the 21 hour quarter period the ceiling allows — and leaves 10 not separable, among them 4 of the 7 that the constant-acceleration comparison resolves.  $\beta$  Pictoris is judged by the same criterion. Its two crossings sit at 241.7739 and 241.8594 GHz in blocks 24.4 hours apart,  $106 \text{ km s}^{-1}$  apart, and both fitted drifts are negative,  $-1611$  and  $-679 \text{ Hz s}^{-1}$ , so a single carrier would have had to rise in frequency between the epochs; that needs a mean line-of-sight acceleration of  $1.21 \text{ m s}^{-2}$ , inside the  $3.6 \text{ m s}^{-2}$  ceiling, so the sign settles nothing on its own. The reversal does: the pair requires one in 16.5 hours, 1.3 times faster than the ceiling permits, which is the narrowest margin of the 3 exclusions and the only one inside a factor of two. The same test accepts

the one pair that is a single emitter, the CO(2–1) line of HD 285968 recovered at one stellar-frame frequency 5.0 days apart, 0.6 standard deviations from the prediction. For the 10 pairs it cannot separate, 2 days to nearly two years apart, nothing measurable tells one carrier from two, and only the agreement of their number with chance bears on them.

The 2 attributed crossings of the 15 tested that are not recovered both remain consistent with their attribution. HR 1010’s crossing at  $230.5287 \text{ GHz}$  is  $-12.0 \text{ km s}^{-1}$  from CO(2–1) and is bounded at  $T_{\text{rep}} < 4.6$  against the  $T_{\text{pers}} = 8.4$  real carbon monoxide at the discovery flux would have given, a disagreement of only  $2.1\sigma$  once that window’s noise is measured over the 47 minutes it integrates,  $11.5 \text{ mJy}$ , where its sibling blocks also lie. ALMA J153702653–33192492 is a serendipitous field star with no measured systemic velocity, so its frame is the barycentric one; its crossing at  $230.5392 \text{ GHz}$  is  $+1.6 \text{ km s}^{-1}$  from the same transition and the bound there is  $T_{\text{rep}} < 4.2$  against  $T_{\text{pers}} = 6.2$ , or  $1.4\sigma$ . Its attribution rests on a velocity rather than on this test: the sightline carries catalogued foreground emission at  $+4.9 \text{ km s}^{-1}$  in the local standard of rest,  $0.5 \text{ km s}^{-1}$  from the crossing’s own velocity in that frame (Appendix C). What its repeat blocks show is where that gas lies rather than whether it can be recovered. All 6 were taken in the same 7 m configuration with synthesised beams of 4.8 to 5.3 arcsec, so an extended cloud is recovered alike in each, and in every one the control ring reaches 10.9 to 13.8 while the phase centre reads 4.2 to 4.8 with 193 to 496 of the 512 control positions above it: the gas is bright and it is not at the phase centre. Toward HD 285968 it is, and over the 4 covering blocks the ring reaches only 6.4 to 8.1 against 4.8 to 6.0 at the star. Whether a foreground line recurs at the phase centre is set by where the cloud is bright, which the control ring measures in the same map.

#### 5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements. This experiment searched 60 systems and 402 Class A windows over  $47.7 \text{ GHz}$  of union bandwidth, and could have confirmed a signal in 37 of those systems — those observed again more than a day later at a frequency the first block covered — and in 11 of them more than a year later (Fig. 8). Table A2 prints it beside the larger count that ignores frequency, and retuning is the difference.

An independent execution block is also not an independent epoch. The baselines run from 0.46 hours to 1.8 yr, and for 21 of the 37 crossings carrying an amplitude the nearest covering block falls within a day. Over months and years these tests bound no duty cycle, and for the other 23 systems they bound nothing at all.

#### 5.6.2. The reserved hold-out and the control beyond 40 pc

Two sets of execution blocks were searched with the frozen pipeline and then reserved for calibration (Table A1); both hold data on real targets and both are reported here as searches. The hold-out — 77 blocks, 315 windows, 36 stars and 214 h on source, withheld by a rule fixed before the search finished and sharing no execution block with the census (Appendix B.1) — contains 8 threshold crossings, the strongest at  $T_* = 8.64$ , and

TABLE 5

**CAN TWO CROSSINGS BE ONE CARRIER?** EVERY PAIR OF CROSSINGS TOWARD ONE STAR, IN TWO EXECUTION BLOCKS, WITHIN  $833 \text{ km s}^{-1}$  OF EACH OTHER IN THE STAR'S FRAME; STARS ARE IDENTIFIED BY POSITION, SO THE TWO ARCHIVAL-TAIL CROSSINGS OF HD 14055 PAIR WITH ITS CATALOGUED ONES.  $\Delta\nu$  IS THE STELLAR-FRAME OFFSET AND  $\dot{\nu}_1$ ,  $\dot{\nu}_2$  THE DRIFTS FITTED INDEPENDENTLY IN THE TWO BLOCKS AND CARRIED INTO THAT FRAME. A TRANSMITTER AT CONSTANT LINE-OF-SIGHT ACCELERATION WOULD PUT THE PAIR AT  $\Delta\nu = \frac{1}{2}(\dot{\nu}_1 + \dot{\nu}_2)\Delta t$ , AND *miss* IS HOW FAR IT IS FROM THAT IN UNITS OF THE DRIFT GRID'S RESOLUTION CARRIED OVER  $\Delta t$ ; A DASH MEANS THAT RESOLUTION EXCEEDS THE WINDOW SEARCHED, SO THE COMPARISON HAS NO POWER. THE LAST COLUMN IS THE VERDICT UNDER THE WEAKER CRITERION OF A BOUNDED TRAJECTORY INSIDE THE SAME GRID AND ACCELERATION CEILING: *no* WHERE ONE CARRIER IS EXCLUDED, *open* WHERE THE PAIR IS NOT SEPARABLE, *yes* FOR THE CONTROL ROW. A FITTED DRIFT IS REFINED INSIDE ONE GRID STEP AND MAY FALL BEYOND THE OUTERMOST TRIAL, HERE BY 0.72 OF A  $59 \text{ Hz s}^{-1}$  STEP, SO A VALUE ABOVE MAY EXCEED THE  $12.0 \text{ nHz}$  CEILING OF §4 ALTHOUGH THE SEARCHED GRID,  $11.99 \text{ nHz}$  THERE, DOES NOT.

| Star        | blocks        | $\Delta t$<br>(d) | $\dot{\nu}_1$<br>( $\text{Hz s}^{-1}$ ) | $\dot{\nu}_2$<br>( $\text{Hz s}^{-1}$ ) | $\Delta\nu$ miss one<br>(MHz) ( $\sigma$ ) carrier? |
|-------------|---------------|-------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------------------|
| 61 Vir      | X822/X82f     | 3.00              | +1092                                   | +3221                                   | +755.4 15 open                                      |
| J1537–3319  | X10359/X398   | 123               | –2000                                   | +2390                                   | +223.5 3.6 open                                     |
| AU Mic      | X1838/X319    | 310               | +1440                                   | +2645                                   | –159.0 — open                                       |
| HD 14055    | X2b2ee/X2c3bb | 0.09              | –2390                                   | +4216                                   | +825.2 2447 no                                      |
|             | X24be0/Xd46   | 1.92              | +1134                                   | –3823                                   | +63.0 41 open                                       |
|             | X2c3bb/X3835  | 2.00              | +4216                                   | +1800                                   | –514.6 144 no                                       |
|             | X2b2ee/X3835  | 2.09              | –2390                                   | +1800                                   | +310.6 48 open                                      |
|             | X3835/Xd526   | 607               | +1800                                   | –2563                                   | –555.5 — open                                       |
|             | X2b2ee/Xd526  | 609               | –2390                                   | –2563                                   | –244.9 — open                                       |
|             | X3835/X25c5e  | 649               | +1800                                   | +2611                                   | +423.3 — open                                       |
|             | X2c3bb/X25c5e | 651               | +4216                                   | +2611                                   | –91.3 — open                                        |
|             | X2b2ee/X25c5e | 651               | –2390                                   | +2611                                   | +733.8 — open                                       |
| $\beta$ Pic | X3a90/Xa9df   | 1.02              | –1611                                   | –679                                    | +85.5 132 no                                        |
| HD 285968   | X108f/X5b4e   | 4.99              | –29                                     | –29                                     | –0.002 0.6 yes                                      |

not one of them outranks its own control positions: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at  $+9 \text{ km s}^{-1}$  from CO(2–1), whose 4 later blocks combine to  $T_{\text{rep}} = +6.71$  at the carried drift, the value Table 7 prints, while the strongest single block reaches 5.87 at its own best drift. It is a molecular line, so its recovery is the test working in the positive direction here as well; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 7).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, with 8 crossings whose strongest is  $T_{\star} = 12.7$  and none of which outranks its own controls. A systemic velocity is known for only 0 of them, so it supports no limit and shows only that the pipeline behaves the same way on fields it was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

### 5.7. A stellar-frame multi-epoch stacked search

Combining the epochs of a star in that star's own rest frame deepens the limit. 55 of the 82 surveyed systems hold more than one searched execution block, and 58 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star's best single epoch and 4.45 at most. 11 further stars stack too — 10 control stars beyond 40 pc and 1 covered only by archival blocks — and are excluded from those figures; over all 69 stacked

TABLE 6

**THE TWO EVENTS OF FIG. 7**, THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ, THE REMAINDER OF THEM COMING FROM THE ARCHIVAL EXTENSION.  $T_{\text{pers}}$  IS WHAT A CARRIER PERSISTING AT THE DISCOVERY AMPLITUDE  $S$  WOULD HAVE GIVEN THERE AND  $T_{\text{rep}}$  IS WHAT IS MEASURED, COMBINING EVERY REPEAT EPOCH BY INVERSE VARIANCE AT THE CARRIED DRIFT: ONE CELL AND ONE TRIAL, SO IT MAY BE NEGATIVE, WHILE THE TWO INDENTED READINGS BENEATH IT ARE MAXIMA AND SO POSITIVE ON AVERAGE WHATEVER IS THERE. THE NEAREST BLOCK ALONE FOLLOWS, BECAUSE THE COMBINATION REQUIRES THE CARRIER TO RETURN TO THAT CELL AT THE START OF EVERY BLOCK ACROSS THE WHOLE BASELINE AND THE SINGLE READING REQUIRES ONLY THE TWO OBSERVATIONS. THE EXCLUSION IS THE FRACTION OF  $S$  STILL RULED OUT AT THE TRIGGER,  $(T_{\text{rep}} + 5)/T_{\text{pers}}$ ; THE DISPLACEMENT PERMITTED IS HALF A CHANNEL IN THE STAR'S FRAME. THE LAST ROW BUT ONE IS THE CHANCE  $\Phi(5T_{\text{pers}}/T_{\star} - 5)$  THAT A CARRIER AT THE TRIGGER AMPLITUDE  $5S/T_{\star}$  WOULD HAVE SATISFIED THE CRITERION, AND IN BRACKETS THE SAME AT  $S$ , WHICH IS SELECTED AS THE LARGEST OF MANY CELLS AND BIASED HIGH.

|                                               | 61 Vir               | CP–72 2713           |
|-----------------------------------------------|----------------------|----------------------|
| Frequency, stellar frame (GHz)                | 346.323236           | 344.296455           |
| as delivered, topocentric                     | 346.313811           | 344.269746           |
| Drift rate ( $\text{Hz s}^{-1}$ )             | +3182                | –2056                |
| $T_{\star}$                                   | 5.958                | 5.809                |
| Highest control $T$                           | 5.65                 | 5.68                 |
| Fitted amplitude (mJy)                        | $118.8 \pm 19.9$     | $11.3 \pm 2.0$       |
| Peak-channel EIRP (W)                         | $5.1 \times 10^{14}$ | $8.9 \times 10^{14}$ |
| Repeat epochs measured                        | 3                    | 5                    |
| of them in the primary census                 | 3                    | 1                    |
| census repeats with no                        | 1                    | 0                    |
| retained spectrum                             |                      |                      |
| Nearest covering epoch,                       | –1.97                | +2.05                |
| separation (h)                                |                      |                      |
| covering epochs more than a                   | 2                    | 4                    |
| day away                                      |                      |                      |
| Longest baseline (d)                          | 3                    | 2                    |
| $T$ expected if persistent, $T_{\text{pers}}$ | 9.70                 | 6.30                 |
| $T$ measured, matched drift, $T_{\text{rep}}$ | +0.39                | –0.82                |
| largest single epoch                          | +1.65                | +0.96                |
| maximised over every drift                    | 2.71                 | 3.60                 |
| Nearest block alone, $T_{\text{pers}}$        | 5.28                 | 6.21                 |
| $T_{\text{rep}}$                              | –0.01                | –0.57                |
| Amplitude still excluded,                     | 0.56 $S$             | 0.66 $S$             |
| $(T_{\text{rep}} + 5)/T_{\text{pers}}$        |                      |                      |
| from the nearest block alone                  | 0.95 $S$             | 0.71 $S$             |
| Displacement permitted                        | $\pm 0.211$          | $\pm 0.213$          |
| ( $\text{km s}^{-1}$ )                        |                      |                      |
| a transmitter at 1 au would                   | 0.042                | 0.044                |
| move                                          |                      |                      |
| and over the longest baseline                 | 1.53                 | 1.08                 |
| Recovery chance at $5S/T_{\star}$ (at         | 1.00 (1.00)          | 0.66 (0.90)          |
| $S$ )                                         |                      |                      |
| Disposition                                   | not confirmed        | not confirmed        |

stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median  $8.2 \text{ km s}^{-1}$  between the epochs of one star and by as much as  $53.9$  — many channels at the  $488 \text{ kHz}$  Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity rather than the directory the search ran in. Spectra with pathological archived visibility weights

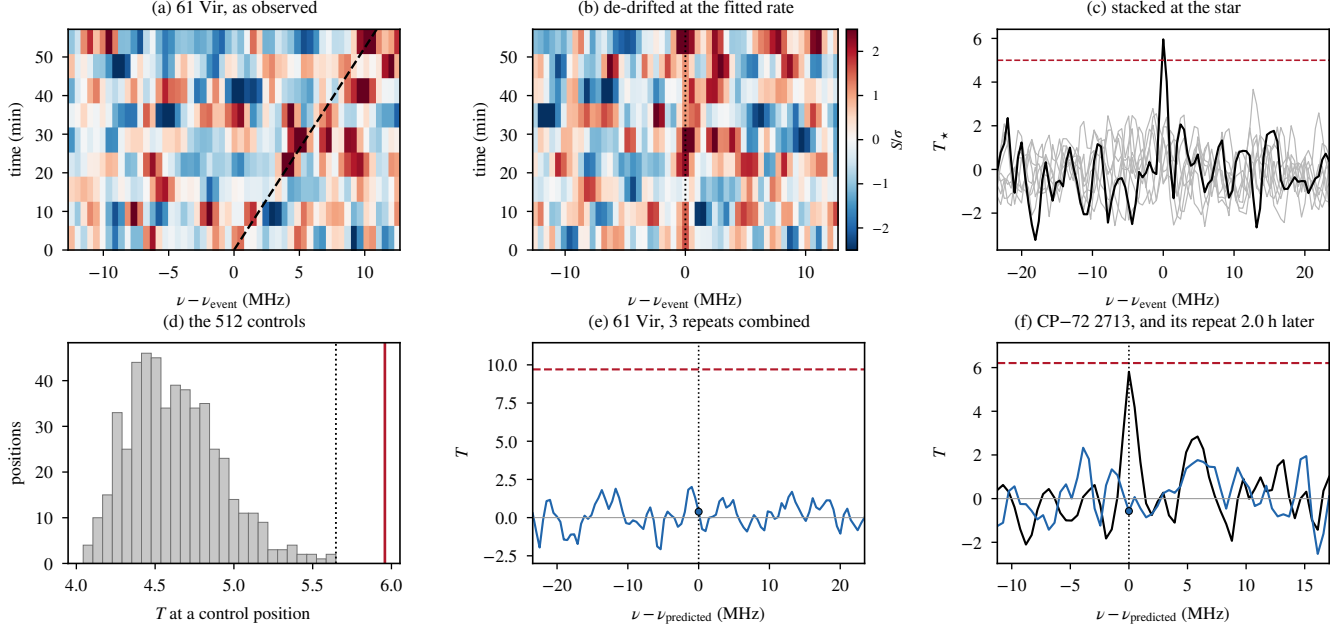


FIG. 7.— **The two crossings that lead their own control fields, and why neither is a confirmed signal.** Frequencies are in the star’s rest frame. (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.3232 GHz, observed 2017 May 20, with the fitted drift of  $+3182 \text{ Hz s}^{-1}$  as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At  $119 \pm 20 \text{ mJy}$  against  $303 \text{ mJy}$  per integration the carrier is 0.39 standard deviations in any one of them, and exists only in the coherent stack of all 38 minutes on source. (b) The same rows shifted by that drift, which carries the carrier 22 channels of  $488.3 \text{ kHz}$  across the observation. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). (e) The 3 other blocks covering that cell, combined, with zero at the predicted channel. Dashed, the  $T_{\text{pers}}$  a carrier persisting at the discovery flux would have given; circled, the  $T_{\text{rep}}$  read there at the carried drift. (f) The second event, CP-72 2713 at 344.2965 GHz: its discovery stack (black) and the same cell 2.05 hours later in a block 1.1 times deeper (blue), with the same two marks, the second at  $T_{\text{rep}} = -0.57$  in that one block and not the combination over all 5 covering blocks. Panels (e) and (f) read one channel at one drift. Table 6 prints every value marked here, and the implied isotropic-equivalent powers of  $5.1 \times 10^{14} \text{ W}$  at  $8.5 \text{ pc}$  and  $8.9 \times 10^{14} \text{ W}$  at  $36.7 \text{ pc}$ .

TABLE 7

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS.  $\Delta v_*$  IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR’S OWN FRAME AND  $n_{\text{rep}}$  THE NUMBER OF COVERING REPEAT BLOCKS;  $T_{\text{pers}}$  AND  $T_{\text{rep}}$  ARE AS IN TABLE D1. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

| Star       | block           | $\nu_{\text{topo}}$ (GHz) | $T_*$ line                                | $\Delta v_*$ | $n_{\text{rep}}$ | $T_{\text{pers}}$ | $T_{\text{rep}}$ | disposition                      |
|------------|-----------------|---------------------------|-------------------------------------------|--------------|------------------|-------------------|------------------|----------------------------------|
| HD 285968  | Xdaef5a_Xdd9b   | 230.5081                  | 8.642 CO(2–1)                             | +8.7         | 4                | 11.6              | +6.71            | attributed; present again        |
| HD 14055   | X107376c_X6126  | 330.7123                  | 5.513 $^{13}\text{CO}$ (3–2)              | +112.6       | 30               | 312.0             | +0.56            | unattributed; does not recur     |
| TRAPPIST-1 | Xd0e450_X259b   | 346.2454                  | 5.326 SO(9 <sub>8</sub> –8 <sub>7</sub> ) | -306.6       | 5                | 11.1              | -0.32            | unattributed; does not recur     |
| HD 172555  | X107fa96_X1300f | 344.6066                  | 5.144 SO(8 <sub>8</sub> –7 <sub>7</sub> ) | +251.3       | —                | —                 | —                | unattributed; no repeat coverage |
| HD 285968  | Xdb4b9a_Xa7c    | 229.9129                  | 5.106 CO(2–1)                             | -768.0       | 4                | 11.2              | +0.21            | unattributed; does not recur     |
| HR 1010    | Xc6b674_X23f2   | 346.2301                  | 5.044 SO(9 <sub>8</sub> –8 <sub>7</sub> ) | -238.7       | 7                | 25.9              | -0.80            | unattributed; does not recur     |
| HD 10647   | Xff99e1_X9cdf   | 345.7059                  | 5.017 SO(2 <sub>3</sub> –2 <sub>1</sub> ) | +37.1        | 13               | 142.1             | -0.93            | unattributed; does not recur     |
| HD 170773  | Xff294f_X1fb    | 345.5597                  | 5.001 SO(2 <sub>3</sub> –2 <sub>1</sub> ) | -113.8       | 4                | 59.3              | +1.45            | unattributed; does not recur     |

are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968’s molecular line collapses from a 19.26-channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked  $Z \geq 6.0$  above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test’s false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is  $\sqrt{N_{\text{eff}}}$  and not  $\sqrt{N}$ , where  $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$  is the equivalent number of equal-depth epochs and the median  $N_{\text{eff}}/N$  is 0.65. What a stack

realises is  $\sqrt{N_{\text{eff}}}$  divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18. Against  $\sqrt{N}$  the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads. HD161868 stacks 13 epochs for a gain of only 1.15: its  $N_{\text{eff}}$  is 1.7, so  $\sqrt{N_{\text{eff}}}$  is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is *Proxima Centauri*’s, and it is a *Class B* limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the

reserved hold-out, all observed over the census blocks' own years. Stacking the census blocks alone, the set every other limit here uses, gives  $\text{EIRP}_{90} = 2.2 \times 10^{13} \text{ W}$ ; adding the rest of the archive deepens that by 1.26 to  $1.73 \times 10^{13} \text{ W}$ , against  $5.51 \times 10^{13} \text{ W}$  for the best single epoch on the same injection-calibrated convention. Its  $N_{\text{eff}}$  is only 7.9, so  $\sqrt{N_{\text{eff}}} = 2.80$  of the measured 3.19 is what combining the epochs buys, the rest being this spectrum's own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel,  $20.9 \text{ km s}^{-1}$  peak to peak, with no inter-epoch drift above  $7.8 \times 10^{-2} \text{ Hz s}^{-1}$ . At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

*2 groups pass the full criterion and both are known lines.* They are  $\beta$  Pictoris' CO  $J = 1-0$  at  $+3.09 \text{ km s}^{-1}$  over 8 epochs ( $Z = 45.3$ ) and its  $J = 2-1$  at  $+3.55 \text{ km s}^{-1}$  ( $Z = 12.1$ ), agreeing to  $0.46 \text{ km s}^{-1}$  or 0.73 of a CO(1-0) channel: the astrophysical positive control passing in the stack. The survey's own  $\pm 50 \text{ km s}^{-1}$  mask leaves 0 of them unattributed. HD 285968 reaches  $Z = 12.89$  on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is extended, as a cloud is and a transmitter is not.

Every stacked limit above holds only for a carrier that stays inside one channel. At CO(2-1) one 488 kHz channel spans  $0.63 \text{ km s}^{-1}$ , the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, its line-of-sight velocity swinging by  $\pm 30 \text{ km s}^{-1}$ , crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to  $1.3 \times 10^{-1} \text{ Hz s}^{-1}$  over the median inter-epoch span of 44 d. No crossing of  $\$5.6$  drifts that slowly: the slowest measured is  $519 \text{ Hz s}^{-1}$ , which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of  $\$6.3$  apply generally.

## 6. DISCUSSION

### 6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth: the power a transmitter must radiate to be found here remains a median  $1.6 \times 10^{15} \text{ W}$  per system.

*Confusion, and not interference, sets the false-positive rate.* Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, arising in 2 of the 69 debris-disc systems searched. The remedy that works is a velocity-space

mask evaluated in the star's frame and in the local standard of rest, and it is not free: it removes 6.4 per cent of the Class A union bandwidth.

*Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths.* A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest.  $\beta$  Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its resolved disc emission lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is  $\times 2.464$  the interior rate; without that window  $\times 1.020$ , and on a held-out half  $\times 1.181$  ( $p = 0.24$ ). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

*The channelisation is inherited.* 1249 of the 1651 windows have channels so coarse that every drift in the grid stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

*Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally.* Let the confirmation coverage  $C_{\text{conf}}(\nu)$  be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency  $\nu$ , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.7 GHz of union bandwidth, 52 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 26 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

### 6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment. Earlier searches above about 30 GHz reached nearby stars at one frequency only, and the one earlier millimetre search of comparable bandwidth worked in Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including  $\beta$  Pictoris' CO  $J = 1-0$  control. Of the 10 MHz–115 GHz haystack axis of Wright et al. (2018) this experiment covers 19.9 GHz, only 1.0 GHz of it carrier search because the Class A union begins at 114 GHz; the rest lies above that axis.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives in-

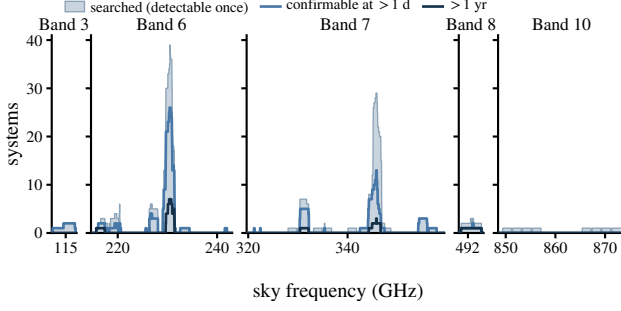


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

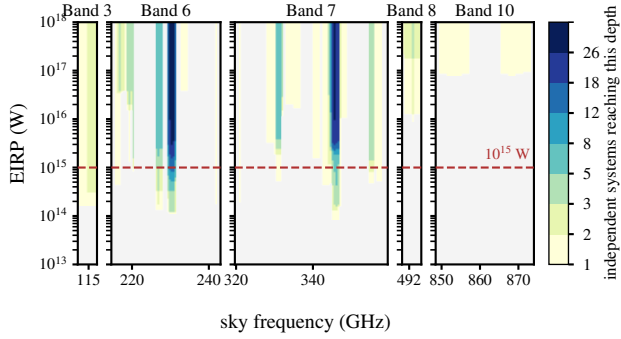


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems in which an unresolved carrier of the plotted EIRP at the plotted sky frequency would have been recovered nine times in ten in a single epoch; unshaded means none. That is a recovery completeness and not a probability of confirmation: confirmation requires a second independent epoch covering the same cell, which Figure 8 maps and which no amount of transmitted power can supply. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is  $1 \times 10^{15}$  W.

stead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of  $1 \times 10^{15}$  W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 39 of the 60 systems, and half the sample only above  $3.4 \times 10^{15}$  W.

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017),  $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$  with  $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$ , smaller being better. Those programmes quote  $\text{EIRP}_{\text{min}}$  at a bare detection threshold, and not at the same one: Enriquez et al. (2017) quote at  $S/N = 25$  and Margot et al. (2023) at 10, where this search triggers at five (Table 1). The measure is linear in  $\text{EIRP}_{\text{min}}$ , so putting every programme on the common footing of five moves the published values — 0.85 becomes 0.17 and 0.14 becomes 0.072 — and it runs against this survey, making the comparison programmes deeper than their own papers report. Margot et al. (2023) publish no CWTFM; that value is the same

expression on the three quantities they do give, 11680 stars, 1.15–1.73 GHz and  $\text{EIRP} > 1.35 \times 10^{13}$  W. This experiment takes no such factor, and on that common footing, with  $N_{\star} = 60$  and  $\nu_{\text{rel}} = 0.097$ , the Class A experiment is  $1.0 \times 10^5$  on the sample’s largest threshold power and  $4.4 \times 10^3$  on the median system, a gap of  $5.2 \times 10^3$  from the published Enriquez value that widens to  $2.6 \times 10^4$  once that value is rescaled with the rest. On the  $\text{EIRP}_{90}$  adopted everywhere else here the same two figures of merit are  $3.2 \times 10^5$  and  $1.3 \times 10^4$ . Nearly all of the difference is the product  $N_{\star} \nu_{\text{rel}}$ : this is a far smaller haystack than a wideband centimetre survey of thousands of stars, and the measure compares surveys rather than estimating occurrence rates.

### 6.3. What this experiment excludes, and what it does not

*The signal these limits describe.* Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and changing frequency linearly at no more than 12.0 nHz. Such a carrier would have produced a crossing nine times in ten above the  $\text{EIRP}_{90}$  of §5.1, a median of  $1.6 \times 10^{15}$  W per system on its best window, and none survived toward any of the 60 systems the carrier search covers. That much holds for every one of them. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 9), the median system covered over only 1.84 GHz; the drift grid reaches an acceleration that contains every known planet of the sample but one (Fig. 3) but not a transmitter on a rapidly rotating body, whose equatorial rotation alone contributes  $2.2 \text{ ms}^{-2}$  for a Jupiter-sized planet, 61 per cent of that ceiling; emission broader than one channel is removed with the spectral baseline; and a carrier on for part of a block comes back at that fraction of its amplitude, while one absent throughout is not constrained at all.

*Recurrence applies over a far smaller domain.* A crossing is not a detection (Table 3), and where the archive observed only once no power makes one. Recurrence could be tested at all for 39 of the 60 systems, and at a frequency the first epoch covered for fewer still (§5.6.1); for the other 21 the confirmation completeness is zero. Where it does apply it excludes a carrier that returns to the cell the first epoch predicts and radiates through both observations, at the amplitudes §5.6 quotes and no weaker ones, over the frequency interval it states and no wider one.

*The result is a non-detection and not an exclusion of persistent transmitters.* The recurrence test reads a single cell at the frequency and drift the first epoch predicts, so the transmitter it tests is one that comes back to that cell, and five restrictions on that model are ours. A transmitter operating intermittently need not have been radiating during the second block. A second block need not cover the frequency at all, and over about half the searched bandwidth none does (§6.1). Orbital motion moves the carrier in the star’s own frame, and the interval the test permits is half a channel,  $\pm 0.08$  to  $\pm 0.63 \text{ kms}^{-1}$  with a median of  $\pm 0.21$ : an orbit at 1 au keeps a transmitter inside it for 14 of the 33 crossings carrying an exclusion over the interval to their near-

est covering block, and for none at the acceleration the drift grid itself reaches. A transmitter may also change frequency deliberately, and nothing here addresses that. And 21 systems have no confirmation epoch of any kind. What this survey supports, therefore, is that no unresolved spectral carrier satisfying the adopted recurrence criterion was confirmed. Outside that criterion nothing is excluded: neither emission inside the  $\pm 50 \text{ km s}^{-1}$  molecular-line mask at any strength, nor any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

*An illustrative engineering benchmark.* A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain  $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$  and so an EIRP of  $5.9 \times 10^{14} \text{ W}$ , a depth this survey reaches for 16 of the 60 Class A systems. The same gain makes its beam  $\Omega = 4\pi/G = 2.1 \times 10^{-8} \text{ sr}$ , 34 arcsec across and illuminating  $1.7 \times 10^{-9}$  of the sky, so detection needs the Earth inside it. EIRP is not transmitter power — the same megawatt radiated isotropically would be  $10^6 \text{ W}$  — so these limits apply to beamed transmitters whose beam contains the Solar System.

*Two blind spots.* The search forms Stokes  $I$  as  $(XX + YY)/2$ , which retains the full intensity of a circularly polarised carrier; what is lost is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission, because ALMA’s feeds are linear and the circularity lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice; none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band.

#### 6.4. What a dedicated ALMA technosignature experiment should do differently

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1249 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 3.05 GHz of the 47.7 GHz Class A union that the mask costs, 6.4 per cent of it — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning:* two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

## 7. CONCLUSIONS

1. The two earlier searches to reach nearby stars above 30 GHz covered a single line about 35 MHz wide at one epoch (§2). What is new here is the bandwidth, 47.7 GHz in 26 intervals, a grid of drift rates rather

than a fixed frequency, and a test for recurrence; and an archive assembled for other science is at present the only way to reach that combination. The one ALMA band below those searched that holds any data on these stars is proprietary until 2027 and channelised far too coarsely to carry a carrier search.

2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in 26 disjoint intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. EIRP<sub>90</sub> is a measured completeness of one epoch and not a threshold: the power at which nine of ten carriers injected into the real calibrated visibilities come back through the unchanged trigger and localisation at the star, inside a single execution block, for a carrier narrower than that window’s own channel. A carrier spread over many channels meets a different limit entirely. The median system’s best window reaches  $1.6 \times 10^{15} \text{ W}$ , with a spread across systems of  $8.7 \times 10^{13}$ – $3.7 \times 10^{16} \text{ W}$ , and 24 systems reach  $10^{15} \text{ W}$ ; carriers were injected into 49 of the 402 windows and the rest carry the class-average factor, whose measured dispersion moves that median by 2.2 per cent, so a single window’s value is an estimate. One bias runs one way only and is outside that calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, which leaves the median limit optimistic by 1 per cent and the windows above 400 GHz by 8 per cent. No limit quoted here carries that correction, and for the 125 blocks whose phase stability cannot be measured, and whose baselines are systematically the longer, the limit is conditional on those observations’ own calibration performance.
4. All 56 threshold crossings are reported with their dispositions, four of them in the one execution block that fails the data-quality criterion of Appendix D.2 and reported apart from the 52 in quality-passing data. Of the 56, 21 fall within  $\pm 50 \text{ km s}^{-1}$  of a catalogued transition at the star’s own velocity, a coincidence that is reproducible and involves no judgement. The mask costs 6.4 per cent of the Class A union bandwidth. Recording every cell above the trigger, rather than each window’s strongest, adds 21 more: three survive attribution and none recurs, so no disposition changes. Chance, measured from each window’s own control positions and channel grid, predicts 28.9 unattributed Class A crossings over the census against the 32 found there, or 30.8 with the measured star-to-control transfer applied, and resampling the execution blocks allows 12–48. The count is compatible with that empirical control expectation, which is a weaker statement than a demonstration that the crossings are noise: the width of the range is its limit, and an added population of as many as 19 crossings would sit inside it unremarked. The one instrumental mechanism that could raise the statis-

tic at the star and not at its controls is a channel-dependent calibration residual. Measured against the continuum the pipeline records at the extraction position, dust within the beam included, 9 windows toward 2 stars admit one inside the documented range, needing 3.8–13 per cent; in all of them the largest statistic at any channel and drift is 3.1 and none holds a crossing, which puts any such residual below 6.6 per cent.

5. Every unattributed crossing was tested for recurrence in the star’s own frame, over the whole of each further window covering the event frequency and at every drift searched, because an accelerating carrier need not return to one channel. None was found again: the largest statistic at any predicted cell is 4.71 against  $T_{\text{trig}} = 5$ . The same criterion recovers 13 of the 15 line-attributed crossings tested, which is the positive control of the stage. Of the 33 crossings that carry an exclusion, the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude with individual limits from 0.017 to 1.00, so the median is not a bound on the set; combining every covering block assumes a carrier that returns to one stellar-frame channel at the start of each, and where the repeat is shallow the criterion has little power and its silence is not an exclusion. Judged at the trigger amplitude, rather than at a discovery amplitude that selection biases high, 4 of these crossings are untested rather than excluded, among them one of the two that lead their own control fields. What these tests exclude is a carrier that holds its stellar-frame frequency to within  $\pm 0.08 \pm 0.63 \text{ km s}^{-1}$  of the first epoch’s and radiates through both observations. An orbit at 1 au keeps a transmitter inside that interval for 14 of the 33 crossings over the interval to their nearest covering block; at the acceleration the drift grid itself reaches, none of the exclusions survive. The conclusion this supports is that no unresolved spectral carrier satisfying the adopted recurrence criterion was confirmed, and not that persistent transmitters are excluded above a power threshold.
6. What this survey could search and what it could confirm are different extents. Of the 60 systems the carrier search covered, 37 were observed again more than 1 day later at a frequency the first block covered; for the other 23 confirmation was beyond reach at any sensitivity, and about those systems the null reported here says nothing at all.
7. Combining each surveyed star’s epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri the stacked limit, measured by injection into the stack, reaches  $2.2 \times 10^{13} \text{ W}$  over the primary census alone and  $1.73 \times 10^{13} \text{ W}$  once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel across 6.3 yr.
8. These results constrain unresolved carriers only within the domain actually searched (§6.3). The drift reach is part of that domain: the grid stops at 12.0 nHz, 6 per cent of the range recommended for a general carrier search (Sheikh et al. 2019), so an un-

bound transmitter, one on a rapidly rotating body, or one whose combined orbital and rotational acceleration exceeds that ceiling is not constrained here at all. What transfers beyond the domain is the obstacle, that channelisation chosen by other observers, and not collecting area, decides which kind of search is possible at all.

#### AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table D1.

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#### COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 July 11 and belong to 65 ALMA proposals, listed in Table A3, whose principal investigators and teams we thank. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

#### DATA AVAILABILITY

Every observation this paper rests on is public. The 404 primary-census execution blocks and the blocks of every other set named in Table A1 are held in the ALMA Science Archive and are retrieved there by the execution-block identifier, which Tables D1 and D2 give for every one of the 56 threshold crossings. The ancillary catalogues are public in the same sense: *Gaia* DR3 for the astrometry and the colours, the NASA Exoplanet Archive

for the planet counts, and the CDMS and JPL line catalogues through Splatalogue for the mask, queried once on 2026-10-06.

The derived results are in the paper. Table A1 partitions the blocks, windows, stars, systems and hours of every set the search defines; Table A2 carries every headline count, the five fates of the unprocessed blocks and the crossing partition; Tables D1 and D2 list all 56 crossings with their frequencies in both frames, the stellar and control statistics, the nearest masked transition and the recurrence test; Table 6 gives the two leading events in full; and Table 7 does the same for the reserved hold-out. The method is set out in enough detail to be reimplemented, and no separate data deposit accompanies the paper.

Three choices that could create or remove an unattributed crossing carry an identifier anyone can check. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as c75069040eab52ec1336376a4dae62114bb281a3 on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a single dated query of a public line catalogue (2026-10-06), so it is specified rather than described, and it post-dates the fixing of the disposition criteria on 2026 September 11 (f5f52e7804f1baad99515ea2d46ddd7e4e4cd546), as does every species in it, so no species in the list was recognised late or early with respect to them.

#### SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

#### APPENDIX

##### A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

This appendix sets out the three things every threshold quoted in this paper is checked from: the frame of each frequency, the instrumental response, and the primary-beam factor that turns a measured noise into a flux at the star.

###### A.1. Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few  $\text{km s}^{-1}$  within 40 pc; second-order relativistic terms at the 50  $\text{km s}^{-1}$  scale are  $\lesssim 5 \text{ kHz}$  at 350 GHz, below the

finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so  $\text{EIRP}_{\text{trig}}$  is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable; and no reprocessing recovers resolution the correlator never recorded. Channel widths come from the measurement sets’ own `CHAN WIDTH`; MS v2 defines no “EFFECTIVE BANDWIDTH” column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is  $\sim 2$  channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are  $\rho_1 = 2/3 = 0.667$  and  $\rho_2 = 0.167$ , with  $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$ : the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier’s flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase.

*That loss enters the limits exactly once, and it is the thresholds rather than the limits that are free of it.* A window’s trigger flux is the bare 5 $\sigma$  of its own spectrum, so setting the *total* power of a channel-confined carrier against it would be optimistic by  $\times 2.00$ – $\times 2.36$ . The injected carriers are not set against it in that way: each is deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh, and recovered through the same trigger, which reads the peak channel. The measured completeness therefore already contains the loss, and factorises into it exactly:  $3.03 P_{\text{trig}}$  is  $\times 2.29$  for the channel response and  $\times 1.33$  for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. So  $P_{\text{trig}}$ , the sensitivity  $S_{\text{min}}$  below and the filled symbols of Fig. 1(a) are nominal throughout, which is what makes them commensurable with other programmes’ detection thresholds, while  $\text{EIRP}_{90}$  carries the correction and is not corrected again. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much; and where in a channel a carrier was placed makes no difference to what is recovered, because at a typical searched drift rate it crosses about 11 channels during one observation and samples the whole response.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so those spectra are smoothed and un-averaged; and the archive’s reported effective resolution is exactly twice the channel separation in 1505 of the 1651 windows, the other 146 reporting a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. *The correlation model is truncated at lag two*: the baseline

step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the retained spectra carry no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

*Transmitters wider than one channel.* The baseline filter (§4.1) is flat below roughly half its effective width  $W$  and falls to zero above it, so  $W$  is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is  $\approx 32$  MHz at 488 kHz and  $\approx 0.9$  GHz at 15.62 MHz. Below it a top-hat transmitter spanning  $N_{\text{ch}}$  channels divides its flux between them at unchanged noise, so its threshold is  $N_{\text{ch}}$  times the single-channel one.

*Intra-integration smearing, which the injections do not model.* Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one. A carrier moving  $\Delta$  channels during an integration keeps  $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$  of its peak amplitude, and every limit in this paper is divided by it. The  $\Delta$  that sets a ninety per cent point is the one at the ninetieth percentile of each window’s own drift draw, 89 per cent of the searched ceiling over the campaign’s 10 332 tones. The integration length is recorded for 1046 of the 1651 windows, 451 of them in the delivered meta-data and 595 more in the exposure records the search kept; for the other 605 the factor is *bounded* at 11.6 s, the longest integration delivered anywhere, where none falls below 0.997. Nothing moves in the median window, and 13 windows on 6 stars rise by more than one per cent, up to  $\times 1.53$ : HIP 10679, HD 172555,  $\beta$  Pic,  $\eta$  Crv, CP–72 2713 and HD 48370. Re-scoring the 49 injected windows’ own ladders with every tone de-rated by its own  $\eta_{\text{smear}}$  gives a penalty never larger than the one applied and smaller by at most 8 per cent, so the correction errs the safe way.

#### A.2. The primary-beam response

The sensitivity quoted for a window is  $S_{\text{min}} = 5\sigma/A$ , where  $A$  is the primary-beam response at the star’s offset from the phase centre: the noise  $\sigma$  is measured in apparent flux, the limit is quoted in true flux, and the two differ by  $A$ . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 windows the offset has a median of  $0.089''$ , a 99th percentile of  $7.43''$  and a maximum of  $8.13''$ , and  $A$  runs from 0.999982 in the median down to 0.5005, below 0.99 on 91 windows and below 0.9 on 38. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most  $\times 2$ , since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap, so no window lies near it and the value chosen does not matter. Beyond it lie the 12  $\epsilon$  Eridani Band 6 windows, withheld as a band rather than window by window. 8 of them, in 2 blocks, need a correction of up to  $\times 3.5$  — past the first sidelobe transition, where the Gaussian form fails — and so fail the floor; the other 4, in block Xb9c01c\_X2004, place the star at the pointing centre and need none, and at 3.22 pc it is too near to be left

without a limit, so theirs are given here: 0.10 window-hours, no crossing, largest statistic 3.8 against a trigger of five,  $1.8 \times 10^{14}$  W in the one fine window at 488 kHz and  $9.3 \times 10^{14}$ – $1.1 \times 10^{15}$  W in the 3 coarse ones at 15.6 MHz. None enters a census statistic. The flux-scale comparison of §4 is a separate continuum measurement on the same star, a block mean over the 4 compact-array executions that place it at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

*The correction is applied exactly once.* The offset is computed for every measurement set from its own FIELD: :PHASE\_DIR and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only:  $S_{\text{min}} = 5\sigma/A$  closes on all 1651 catalogued rows to  $8.7\text{e-}05$  in relative terms. On 10 rows whose products are not retained  $A$  is back-solved from  $5\sigma/S_{\text{min}}$  and flagged as such, so there the closure is arithmetic; the test is the other 1641 rows, where  $A$  comes from the search’s own stored geometry and closes to  $8.7\text{e-}05$ . Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which  $A$  divides out identically.

*The response applied is a Gaussian of ALMA’s own primary-beam full width,  $1.13\lambda/D$*  (Cortes et al. 2026), evaluated at each window’s *own* dish diameter. The dish matters: the 1054 rows taken on the  $1.7\times$  smaller compact-array antennas have a correspondingly wider beam, and a Gaussian forced to  $D = 12$  m reproduces the per-window geometry only to  $7.7 \times 10^{-2}$  against  $2.6 \times 10^{-5}$  for the per-dish form. The wider Airy first-null coefficient  $1.22\lambda/D$  is sometimes used in its place; it attenuates less, and so would make every limit deeper, by 0.001 per cent in the median Class A window and 0.02 per cent in the median system, by more than one per cent in 40 of 1651 windows and by more than ten in 4 — those few all LP 736-15 at  $8.1''$  off axis, the largest being 10.3 per cent. At ALMA’s width the lowest response in the sample is 0.5005 rather than 0.5522, still clear of the 0.5 retention floor, so the sample and every disposition in it are the same under either choice. The remaining model uncertainty is bounded by measurement: CASA’s blocked-Airy response is better physics than a Gaussian, the largest ratio to it over the whole sample is 1.0745, and the counts of systems reaching  $10^{15}$  W are identical under any of the three.

*The star’s apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled.* A small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines, so the size of the displacement decides whether a scalar correction is legitimate at all. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax and radial velocity where catalogued — leaves each window’s noise unchanged to 0.20 per cent and alters no disposition, but it cannot test the amplitude factor, because that factor describes a source and those windows hold none: moving the extraction point by two thirds of a beam simply draws a fresh noise realisation, and the statistic scatters from 0.78 to 1.15 about 0.98 instead of approaching the 0.49 a source would give. A

carrier injected at the apparent position and recovered from the *same* visibilities at both positions does return the normalised dirty beam at the omitted displacement, on that block’s own baselines and with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. That it exceeds the model at the larger displacement is the search’s own doing, since maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it need be.

*The injection campaign is blind to  $A$  by construction, and the bias that follows is exactly zero.* The injector deposits an unattenuated amplitude at the star’s own direction cosines and refers the ladder to the *apparent*  $5\sigma$  on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by  $1/A$  while  $P_{\text{trig}}$  is the power of a true  $5\sigma/A$  and  $A$  cancels identically; 4 of the units have apparent and corrected  $S_{\text{min}}$  differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches out to an offset of  $4.98''$ , at which the response is 0.9102, whereas 75 of the 1651 released windows, on 5 stars, lie further out than any injected one, down to  $A = 0.552$ , so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected. That set is defined by offset and is not the 38 windows whose response falls below 0.9, which is a different criterion and a smaller set.

*A few rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled:* on 4 windows of HD 285968 in block A002\_Xd98580\_X3bf6 the catalogued  $\sigma$  is 0.88–1.32 per cent *higher* than the surviving product’s, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained the declared provenance gap is 14 windows.

## B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; were it operating it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, with no free parameter.

*Applied out of sample, the correction does not repair*

*the statistic.* Frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately; rescoring the whole survey with the gradient removed moves the crossings outranking all their controls from 12 windows to 11, and leaves the survey median rank slightly further from one half than before. The gradient is real but the residual is not radial, and neither normalisation yields a calibrated tail probability, so we keep the frozen rank and rest every disposition on physical evidence or on recurrence. Because the rank gates nothing, none of this enters a sensitivity limit.

*A calibration error that varied with frequency would explain the displacement, and it is the only mechanism on offer.* An error identical in every channel is removed exactly by the per-integration median along frequency; a channel-dependent one is not, and two ordinary errors are channel-dependent: a residual in the bandpass, and a system temperature measured in the coarse correlator mode and interpolated onto the science channels, which near a telluric line leaves a narrow feature behind (Hunter et al. 2018). Both multiply the continuum, so both act at the star and not on empty sky, both stand still in topocentric frequency and both are coherent between integrations: they would stack at near-zero drift and put the stellar rank *below* one half, the sign observed.

*Amplitude alone does not exclude them.* What such an error multiplies is the whole continuum at the extraction position, a dirty-map amplitude that includes any dust inside the synthesised beam; what it must beat is the noise of the de-drifted, block-combined spectrum, since a coherent residual stacks as a carrier does. A fractional error  $\epsilon$  deposits  $\epsilon S_c$  at the star, so a crossing at the trigger needs  $\epsilon = 5\sigma/S_c$ . The survey’s own continuum maps — 7 of them, each used only on windows of its own band — give 63 mJy at L 836-122, a  $66\sigma$  detection and a factor  $1.3 \times 10^4$  above that star’s photosphere; on the measured continuum 9 windows of 2 stars need only 3.8–13 per cent, inside the 5–20 per cent a coarse system temperature is known to leave at an ozone line; and at AU Mic, where the same extractor measures 29.2 mJy near 106 GHz, 1.4 per cent would suffice.

*Those windows measure the residual instead, and it is too small.* In all 9 of them the largest statistic anywhere — every channel, every drift — is  $T_\star = 3.1$  where the trigger is five, the largest control probe 5.4, and none holds a crossing. Read as a fraction of the continuum beside it, the coherent channel-dependent residual at the extraction position is below 6.6 per cent in all of them and below 1.9 per cent at L 836-122, whose continuum was measured in the one execution block its windows come from — beneath the 5 per cent floor of the documented range. Their median stellar rank, 0.61, lies on the far side of one half from the displacement the mechanism is invoked to explain. CD-57 1054 carries the whole pattern within one star: 6 of its windows would admit 9–13 per cent and none produces anything, while its one crossing lies in the window where 71 per cent would be needed. The crossings are not where the continuum is: 52 of the 56 are in fine-channel windows, where this route needs 36–4115 mJy within one beam of the star and only

TABLE A1

**THE SURVEY LEDGER, PART ONE: EXECUTION BLOCKS AND WINDOWS.** EACH ROW CARRIES ITS OWN NUMBER, AND A GROUP SUMS TO THE ROW WHOSE NUMBER HEADS IT — ROWS 3.1.1 TO 3.1.4 TO ROW 3.1, ROWS 3.1 TO 3.5 TO ROW 3. THE PRIMARY CENSUS IS THE SURVEY; UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE WINDOW-HOURS: ON-SOURCE TIME SUMMED OVER SPECTRAL WINDOWS, SO THE CLASSES PARTITION THE HOURS AS THEY PARTITION THE WINDOWS. THE TELESCOPE ON-SOURCE TIME BEHIND THE PRIMARY CENSUS IS 280 h, OF WHICH 152 h CARRIES AT LEAST ONE CLASS A WINDOW AND 276 h AT LEAST ONE CLASS B WINDOW — FIGURES THAT OVERLAP AND MUST NOT BE ADDED. CONFIRMED PLANET COUNTS HERE AND IN THE TARGET LIST COME FROM THE NASA EXOPLANET ARCHIVE **PSCOMP** TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4. PART TWO, TABLE A2, CARRIES THE EVENTS AND THE CONFIRMATION EXTENTS.

| Set of execution blocks                                | Blocks | Windows | Stars | Systems | Hours | Results it enters                                                              |
|--------------------------------------------------------|--------|---------|-------|---------|-------|--------------------------------------------------------------------------------|
| 1 Archival scope: blocks the query exposes             | 656    | —       | —     | —       | —     | rows 1.1 and 1.2                                                               |
| 1.1 ... processed with the frozen pipeline             | 479    | —       | —     | —       | —     | panel 2                                                                        |
| 1.2 ... not processed at the freeze                    | 177    | —       | —     | —       | —     | panel 3                                                                        |
| 2 Processed with the frozen pipeline <sup>c</sup>      | 484    | —       | —     | —       | —     | rows 2.1 to 2.4                                                                |
| 2.1 <b>Primary census</b>                              | 404    | 1651    | 89    | 82      | 1094  | every limit, the crossing list, every chance expectation                       |
| 2.1.1 Class A, fine channels <sup>a</sup>              | —      | 402     | 65    | 60      | 227   | the drifting-carrier experiment                                                |
| 2.1.2 Class B, coarse channels <sup>a</sup>            | —      | 1249    | 88    | 81      | 867   | the spectral-excess experiment                                                 |
| 2.2 Reserved hold-out                                  | 77     | 315     | 36    | —       | 214   | the rank displacement, the false-alarm tail factor, the radial control profile |
| 2.3 Withheld: primary-beam form invalid <sup>d</sup>   | 2      | 8       | —     | —       | —     | nothing; withheld                                                              |
| 2.4 $\epsilon$ Eri at the pointing centre <sup>e</sup> | 1      | 4       | 1     | 1       | 0.1   | reported limits only; no census statistic                                      |
| 3 Not processed at the freeze: what became of each     | 177    | —       | —     | —       | —     | rows 3.1 to 3.5, the five fates                                                |
| 3.1 Calibrated and searched                            | 144    | —       | —     | —       | —     | rows 3.1.1 to 3.1.4                                                            |
| 3.1.1 <b>Archival extension</b> , reported             | 115    | 463     | 16    | 16      | 269   | recurrence, and the null on data the frozen pipeline had not seen              |
| 3.1.2 Searched after the extension was frozen          | 8      | 32      | —     | —       | —     | nothing; later than the reported extension                                     |
| 3.1.3 Withheld: primary-beam form invalid <sup>d</sup> | 8      | 32      | —     | —       | —     | nothing; withheld                                                              |
| 3.1.4 No window reached the export                     | 13     | 0       | —     | —       | —     | nothing                                                                        |
| 3.2 No calibration in the ALMA delivery                | 18     | —       | —     | —       | —     | nothing; the archival gap                                                      |
| 3.3 Calibration failed                                 | 11     | —       | —     | —       | —     | nothing; the archival gap                                                      |
| 3.4 Not attempted                                      | 3      | —       | —     | —       | —     | nothing; the archival gap                                                      |
| 3.5 Working disk insufficient                          | 1      | —       | —     | —       | —     | nothing; the archival gap                                                      |
| 4 Beyond 40 pc: out-of-sample control                  | 47     | 198     | 22    | —       | 88    | the external null; no science count                                            |
| 5 External calibration sample <sup>b</sup>             | 326    | 1322    | 56    | —       | —     | the rank calibration and the positive control; no science count                |

<sup>a</sup>The two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

<sup>b</sup>Not an out-of-sample set and not an addition to this ledger: every one of its 326 blocks is counted in a row above, 263 in the primary census and 63 in the hold-out, and it is listed apart only to keep it from being counted twice. One further block of the harvest it came from lies outside the 40 pc work list and is excluded here.

<sup>c</sup>Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 3 of the 5 are in the primary census.

<sup>d</sup>Every one is an  $\epsilon$ Eri Band 6 block, and the band was withheld whole: in most of them the star sits far enough off axis (response 0.28–0.35, below the 0.5 floor) that the Gaussian primary-beam form is not valid.

<sup>e</sup>The windows of the same band that place the star at the pointing centre, at a response of 1.00, and so pass the floor; their limits are in Appendix A and they enter no census statistic. Row 3.1.1 holds a further 16 such windows, which enter nothing either.

6 ask less than the brightest continuum this sample has anywhere. The bandpass branch fails by amplitude everywhere, needing  $4\times$  the 1 per cent its worst band is specified to (Kameno 2014); and no crossing of either sample lies within 25 MHz of a catalogued atmospheric transition (Appendix E), against the 31.2 MHz resolution at which the system temperature is measured. One further question the shared noise scale raises is settled by the same arithmetic: over the 1595 windows carrying no crossing the stellar statistic divided by the median of the 512 control statistics — which *is* the ratio of the two noises — has median 1.001 (0.994–1.004).

### B.1. The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if  $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$ , subject to two guards that use identifiers alone — a block on the pre-registered list stays in the survey, and no system loses every block

— so that the assignment follows from the UIDs and that list alone. The statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so no diagnostic built afterwards can promote a window. Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Two quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star outranks all 512 probes in 14 of the survey's windows against 3.2 if the positions were interchangeable, so the direction inflates star-first outcomes rather than suppressing them. Clustering removes neither the shift nor its size: 99 of 149 blocks have a mean rank below

TABLE A2

**THE SURVEY LEDGER, PART TWO: EVENTS, DISPOSITIONS AND CONFIRMATION.** EVERY HEADLINE COUNT IN THIS PAPER IS READ FROM HERE. THREE COUNTS ARE 52 OVER THREE DIFFERENT SETS — THE CLASS A CROSSINGS OF THE ADOPTED LEDGER, THOSE OF THE EXTRACTION CATALOGUE, AND THE QUALITY-PASSING CROSSINGS. THE FIRST TWO HOLD DIFFERENT ROWS, SINCE RE-EXTRACTION MOVED FOUR CROSSINGS ACROSS THE TRIGGER EACH WAY; THE LAST TWO HOLD THE SAME ROWS, BECAUSE THE ONE BLOCK FAILING THE QUALITY CRITERION IS ALSO THE SURVEY’S ONLY BLOCK WITH A COARSE-CHANNEL CROSSING. ROW 3.1.1 IS WHAT THIS PAPER MEANS BY CONFIRMATION AND THE ONLY SUCH COUNT IT QUOTES: A REPEAT EPOCH AT A FREQUENCY THE FIRST DID NOT COVER CONFIRMS NOTHING, SO ROWS 3.1 AND 3.2 COUNT ANY SECOND EPOCH AND ARE NOT CONFIRMATION COUNTS. ROW 2.2.1 WAS ALSO DECOMPOSED SYSTEM BY SYSTEM, EACH EXPECTATION FORMED FROM THAT SYSTEM’S OWN CONTROL POSITIONS: 2 OF THE 60 SYSTEMS FALL BELOW ONE CHANCE IN TWENTY, AGAINST 3 EXPECTED, AND HD 14055, WHICH HOLDS 8 OF THE 56 CROSSINGS, 8 OF THEM UNATTRIBUTED, GIVES 6 IN CENSUS WINDOWS AGAINST 4.7 EXPECTED OVER ITS 38 CLASS A WINDOWS. NO SYSTEM DEPARTS FROM ITS OWN CONTROLS BY MORE THAN CHANCE ALLOWS.

| Count                                               | Value     | What it counts                                                                                      |
|-----------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------|
| 1 Threshold crossings at $T_\star \geq$             | 56        | census and archival tail                                                                            |
| 5                                                   |           |                                                                                                     |
| 1.1 Class A, fine channels                          | 52        | the carrier experiment                                                                              |
| 1.1.1 ... in the primary census                     | 50        | catalogued window by window                                                                         |
| 1.1.2 ... in archival-tail blocks                   | 2         | both toward HD 14055; in no expectation                                                             |
| 1.2 Class B, coarse channels                        | 4         | all in the one block failing §5.3.2                                                                 |
| 2 Quality-passing crossings                         | 52        | the same rows as 1.1                                                                                |
| 2.1 ... attributed to a transition                  | 18        | inside the mask in the star’s own frame                                                             |
| 2.2 ... unattributed                                | 34        | rows 2.2.1 and 2.2.2                                                                                |
| 2.2.1 ... in the primary census                     | 32        | against 28.9 expected by chance, measured at the 402 windows’ own control positions (12–48 allowed) |
| 2.2.2 ... in archival-tail blocks                   | 2         | outside that comparison                                                                             |
| 2.2.3 ... above all 512 controls                    | 2         | the two events of Fig. 7                                                                            |
| 2.2.4 ... recovered in a later block                | 0         | no confirmed signal; 13 attributed ones do                                                          |
| 3 Class A systems                                   | 60        | of 82 surveyed                                                                                      |
| 3.1 ... observed again more than 1 day later        | 39        | at any frequency, so not a confirmation count                                                       |
| 3.1.1 ... at a frequency the first block covered    | <b>37</b> | <b>the confirmable set, and the only one quoted; 11 of them over a year later</b>                   |
| 3.2 ... observed once, or all within one day        | 21        | 18 in a single block, 3 within a day                                                                |
| 3.3 ... with no second block at a covered frequency | 23        | confirmation impossible: row 3.2 and those whose later blocks share no frequency; 36 of all 82      |
| 4 Class A union bandwidth                           | 47.7 GHz  | 26 disjoint islands                                                                                 |
| 4.1 ... removed by the line mask                    | 6.4 %     | 3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union          |

0.5 and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46), while the searched sample leans the same way but by less (median 0.49, 0.47–0.51), the two intervals not overlapping. Either way it runs in the direction that inflates star-first outcomes, which is the quantitative reason the rank is read as a diagnostic and never as a significance.

A median, however, constrains the centre of a distribution, and the crossings are drawn from its extreme tail.

TABLE A3

**THE 65 ALMA PROPOSALS THE SEARCHED EXECUTION BLOCKS BELONG TO, IN THE STANDARD ADS/JAO.ALMA#code FORM.** EVERY BLOCK OF THE PRIMARY CENSUS BELONGS TO ONE OF THESE, AND THE BLOCKS THEMSELVES ARE RETRIEVED FROM THE ALMA SCIENCE ARCHIVE BY THE EXECUTION-BLOCK IDENTIFIERS TABLES D1 AND D2 GIVE.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

The figure that bears on the chance comparison is therefore the rate at the trigger itself, and the same reserved blocks measure it: stars reach the trigger in 8 of 62 fine-channel windows where those windows’ own control positions predict 7.5, a ratio of 1.07 with 0.48–1.79 allowed by resampling the blocks, and carrying it through moves the expectation of Appendix D from 28.9 to 30.8 against the 32 found. That interval is wide — it permits a stellar exceedance rate anywhere from about half the control rate to nearly twice it — so what 8 events on 62 windows establish is compatibility with the empirical control expectation, not the shape of the tail, and a factor of two either way would not show.

Second, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of  $+0.067 \pm 0.033$ , small and of the wrong sign to account for the displacement.

The *external calibration sample* is not an out-of-sample set and must not be confused with the hold-out. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have; 263 are primary-census blocks, 63 hold-out blocks and 0 outside both, so it enters no count of the science sample because its blocks are counted there already. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls, 4 of them  $\beta$  Pictoris CO, and 3 unattributed against 3.7 expected.

## B.2. What the recurrence statistic reads when nothing is there

The exclusion in Table D1 is one formula. Writing  $S$  and  $\sigma_d$  for the discovery amplitude and its error, and  $\bar{a}$  and  $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$  for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\text{excl}} = \frac{S - \bar{a}}{\sqrt{\sigma_d^2 + \bar{\sigma}^2}} = \frac{T_{\text{pers}} - T_{\text{rep}}}{\sqrt{1 + (T_{\text{pers}}/T_\star)^2}}, \quad (\text{B1})$$

with  $T_{\text{pers}} = S/\bar{\sigma}$  and  $T_{\text{rep}} = \bar{a}/\bar{\sigma}$ , so every exclusion follows from three columns of that table and nothing else.  $T_{\text{rep}}$  is read at the carried drift and at one cell; the largest value over every drift trial at the same cell

is tabulated beside it but is never compared with  $T_{\text{pers}}$ , since a maximum is positive on average whatever is there. No exclusion can exceed the strength of the discovery it is built from, nor is it the strongest statement available: an amplitude selected as the largest of  $10^2$  to  $10^6$  cells at  $T_\star \simeq 5$  is biased high, so a real carrier would be fainter than measured, and what survives the bias is the limit  $(T_{\text{rep}} + 5)/T_{\text{pers}}$  — the fraction of the discovery amplitude still ruled out at the trigger.

*The same three numbers say where the test has power and where it has none.* The criterion asks the repeat to reach the trigger, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability  $\Phi(T_{\text{pers}} - 5)$ , and at the trigger amplitude  $5S/T_\star$  — the faintest carrier that could have produced the crossing at all — with  $\Phi(5T_{\text{pers}}/T_\star - 5)$ . Where they are small the crossing is untested rather than excluded, and is reported that way.

*The reading at the predicted channel is centred on zero and scatters by one, measured and not merely designed.* Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is  $+0.014$  and its scatter  $1.046$ , and at the control positions of the discovery windows  $-0.036$  and  $1.034$ ; over the 2 crossings whose repeat spectra survive, the largest value anywhere within  $\pm 15 \text{ km s}^{-1}$  of a predicted channel is  $3.19$ . A maximum behaves as a maximum should: the largest of the 8 retained control probes of a repeat block averages  $1.449$  against  $1.424$  expected, the largest of the  $n_{\text{rep}}$  per-block readings  $+1.102$  against  $+1.130$ , and the combined  $T_{\text{rep}}$  of Eq. B1  $-0.119$ . Widening the window costs trials and not calibration: the exceedances of  $\$5.6$  match the same accounting, and HD 69830’s windows are the only ones too narrow for the excursion their crossing’s drift implies.

### C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It reports the frequency coincidence and the judgement about the coinciding species separately, so the reproducible part can be recomputed and the arguable part argued with.

*The list, and when it was made.* The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN,  $\text{HCO}^+$ , CS, CN, SiO,  $\text{H}_2\text{CO}$ , SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table C1). No strength cut is applied across species: any threshold admitting HCN’s strong lines would discard the brightest lines in these fields. Within a species the main hyperfine components are kept and the satellites dropped. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6, after the disposition criteria were fixed on 2026 September 11, and frozen before any crossing was compared with it.

*Attribution is decided in the star’s own rest frame, and the sign is not the usual one.* A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. That frame, and no other, is where a coincidence

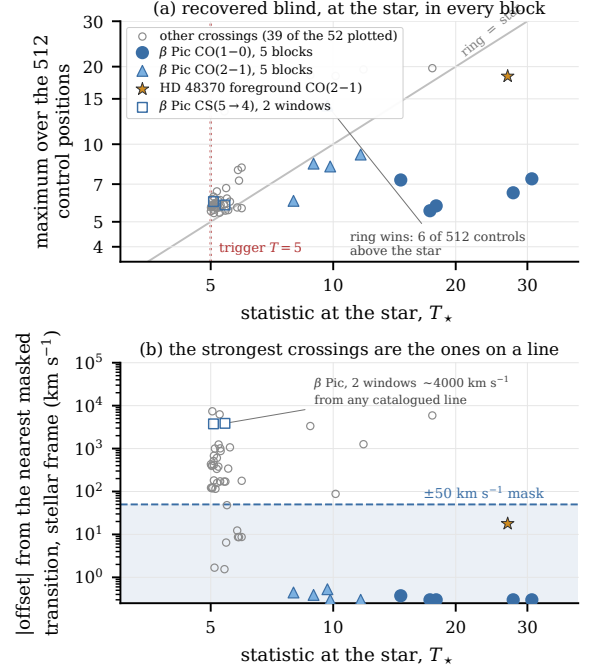


FIG. B1.— The  $\beta$  Pictoris positive control, in both directions. Both panels draw the same 52 of the 56 crossings, the remainder having no retained control ring. (a) The spatial screen: the stellar statistic against the maximum over that window’s 512 control positions. Filled symbols are the blocks in which the blind search recovers circumstellar carbon monoxide at  $\beta$  Pictoris, 9 of them with the star above every control and one with the ring above it, block X7116f1\_X1785, Band 6, where the ring reaches  $14.13$  against  $9.68$  at the star ( $\$5.4$ ). (b) The same crossings by velocity offset from the nearest masked transition in their own star’s rest frame, with the adopted  $\pm 50 \text{ km s}^{-1}$  half-width marked: the 10  $\beta$  Pictoris carbon-monoxide crossings land  $0.1$ – $0.5 \text{ km s}^{-1}$  from CO, and its 2 CS( $5 \rightarrow 4$ ) crossings  $3765$ – $3870 \text{ km s}^{-1}$  from anything catalogued. 39 further crossings are grey, HD 48370 marked.

is judged. The barycentric term alone spans  $58 \text{ km s}^{-1}$  across the survey, so a tolerance applied in the observed frame would spend most of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked:  $12 \text{ km s}^{-1}$  apart as observed, they agree to  $0.03 \text{ km s}^{-1}$  in the star’s frame. Throughout,  $\Delta v_\star = c(\nu_\star - \nu_0)/\nu_0$ , so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

*The local standard of rest is read as well, as a check.* Gas along the line of sight is at rest in that frame rather than in the star’s, so for a star with a large LSR velocity a foreground line could fall outside a stellar-frame mask by construction. Every crossing is therefore carried one further step, the projection of the standard solar motion, and read there too. The two offsets differ by the star’s own LSR velocity and by nothing else, which here runs  $0.3$ – $18.7 \text{ km s}^{-1}$  about a median of  $3.0$ . no crossing is attributed there that the star’s frame does not attribute already, so the check moves nothing; nor could it have, since the unattributed crossing nearest a transition lies  $88 \text{ km s}^{-1}$  from one,  $38 \text{ km s}^{-1}$  beyond the mask and twice the largest of those velocities. One coincidence runs the other way and exists only in the star’s frame, HD 161868’s  $-48.0 \text{ km s}^{-1}$  from SO( $2_3$ – $2_1$ ) against  $-53.4$  in the local standard of rest, which is a further reason to

TABLE C1

THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS  $N$  IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED  $E_u$ . THE CRITERION AND THE DATED QUERY ABOVE DEFINE THE LIST EXACTLY, SO THE INDIVIDUAL TRANSITIONS CAN BE RECOVERED FROM THE LINE CATALOGUES THEMSELVES.

| Species           | $N$ | $\nu_0$ (GHz) | $E_u$ (K) |
|-------------------|-----|---------------|-----------|
| H                 | 5   | 92.03–231.90  | –         |
| SO                | 15  | 100.03–682.68 | 16–143    |
| H <sub>2</sub> CO | 13  | 101.33–491.97 | 10–106    |
| C <sup>18</sup> O | 3   | 109.78–329.33 | 5–32      |
| CN                | 50  | 113.12–680.30 | 5–114     |
| CO                | 3   | 115.27–345.80 | 6–33      |
| CS                | 3   | 195.95–342.88 | 24–66     |
| SiO               | 2   | 217.10–347.33 | 31–75     |
| <sup>13</sup> CO  | 8   | 220.40–330.59 | 16–32     |
| HCN               | 7   | 354.50–354.51 | 43–43     |
| HCO <sup>+</sup>  | 1   | 356.73–356.73 | 43        |
| [C I]             | 1   | 492.16–492.16 | 24        |

decide in the frame circumstellar gas shares.

*What the second reading establishes is which coincidences are foreground.* Of the 24 stars carrying crossings, the composite CO(1 → 0) survey of Dame & Thaddeus (2022), which contains Dame et al. (2001), covers 11 and holds emission toward 5 of them, between  $-18$  and  $+22$  km s<sup>-1</sup>; the all-sky Planck CO map (Planck Collaboration et al. 2016) stays below 0.178 K km s<sup>-1</sup> toward every other, including the 13 outside the CfA footprint. Toward ALMA J153702653–33192492, HD 139084 B, HD 285968 and HD 48370 the CO spectrum is non-zero at the crossing’s own LSR velocity, within 3 km s<sup>-1</sup> of its own peak and at 0.23 K or brighter against a 0.18 K channel: 6 of the 21 coincidences are foreground gas and not circumstellar. Two are worth naming. ALMA J153702653–33192492 is a serendipitous field star toward the Lupus clouds, at 35.0 pc with the clouds far behind it, and its CO(2 → 1) crossing lies 0.5 km s<sup>-1</sup> from a 6.6 K km s<sup>-1</sup> cloud line at  $+4.9$  km s<sup>-1</sup>: the field is a molecular sightline and not a disc. The velocity says so and the field does not: its brightest control position stands above the star in all six of its windows, but so it does in 390 of the 402 Class A windows of the survey, so that comparison distinguishes nothing. Its other 5 crossings lie at least 122 km s<sup>-1</sup> from every transition of the list, and the mask disposes of none of them. HD 48370 is the case where the two readings part: its coincidence lies  $-17.8$  km s<sup>-1</sup> from the star’s own CO(2 → 1), while in the local standard of rest it sits at  $+23.9$  km s<sup>-1</sup>, on a catalogued cloud line peaking at  $+21.8$ . The offset follows the foreground gas and not the star, so that window’s extended emission is foreground cloud and not a resolved disc.

*The coincidences, and the one species we do not read as emission.* Of the 56 crossings, 21 fall within  $\pm 50$  km s<sup>-1</sup> of a transition of the list in their own star’s frame; that count is the first layer and all the velocity criterion decides. Coincidences with sulphur monoxide account for 3 — CD–57 1054, CP–72 2713 and HD 161868, at  $-1.7$ ,  $-12.3$  and  $-48.0$  km s<sup>-1</sup> from transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO(3 → 2) — and two things argue against reading those as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O; and toward CP–72 2713 the CO(3 → 2)

line falls in the same window and is absent to 10.5 mJy in a 0.42 km s<sup>-1</sup> channel while the crossing itself is 12.2 mJy, so an SO line there would have to outshine an undetected CO(3 → 2) line in the same data. Both arguments are about the species and not about any one crossing, so all 3 are treated alike: Table D1 prints the offset beside each with a dagger and disposes of none of them, giving 18 attributed against 38 unattributed, of which two outrank every control position in their own windows. Reading them as attributions instead gives 21 and 35, and 1 such crossing; the recurrence test excludes a persisting carrier for all 3 either way, at  $4.1$ – $5.5\sigma$ , so the choice moves counts and no conclusion. The reserved hold-out is treated identically, leaving 1 attributed and 7 unattributed there (§5.6.2).

*The half-width, and what it costs.* A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive,  $\lesssim 13$  km s<sup>-1</sup> beyond 10 au, and is  $\pm 50$  km s<sup>-1</sup>. At that width the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent of it, against 5.04 GHz of the 118.1 GHz both classes cover, 4.3 per cent: the cost falls on the primary experiment, because its windows are the line-tuned ones. 6.4 per cent is the cost of the mask this paper applies, and is the figure quoted elsewhere. Attributing on either frame instead would mask the union of two tubes separated by the star’s LSR velocity, 3.72 GHz or 7.8 per cent of the Class A union — 22 per cent more bandwidth for no further attribution, which is the measurement behind deciding in one frame. Between  $\pm 20$  and  $\pm 100$  km s<sup>-1</sup> the mask labels 17 to 19 of the crossings, and the 2 that outrank all 512 control positions of their own windows are the same two at every width.

*What a bright line costs is its channels, and nothing beyond them.* Because every above-trigger cell of a window is recorded (§5.3), a weaker carrier beside a molecular line is recorded too, and the price of line confusion is the masked channels and not the windows holding them. Only the 3 fine windows narrower than the mask itself are lost outright, 0.108 GHz, 0.23 per cent of the Class A union; the one masked window that still holds a trigger-level cell outside the mask,  $\beta$  Pic at 115.6200 GHz, does not recur in the 4 other blocks covering that cell.

*The labelled crossings are tested for recurrence too.* 15 of the 18 attributed crossings have a later block covering the predicted cell, and the 13 that are present again, toward  $\beta$  Pic and HD 285968, are the positive control of that stage. For the 2 not recovered a repeat window’s own largest statistic bounds the predicted cell from above and Table D1 prints the row as a bound; 3 have no covering block at all (HD 139084 B, HD 48370 and HIP 10679).

*Image-sideband leakage accounts for no crossing, and a longer list would not be a better one.* The mask holds signal-sideband frequencies, and the 2 unattributed crossings that share a block with a CO(2 → 1) crossing in the opposite sideband would need a first local oscillator outside the range their own block’s spectral windows permit, 235.02–235.68 GHz. In the other direction, every Splatalogue transition over these intervals with an Einstein coefficient above  $10^{-5}$  s<sup>-1</sup> — 5,114 transitions of 235 species — would mask 75.1 per cent of the union and move 27 crossings across the line, one of them rank-leading. At these frequencies almost anything lies within

$\pm 50 \text{ km s}^{-1}$  of something in such a list, which is why the species are restricted to what these setups cover and why the outcome is a label rather than a verdict.

#### D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix gives the chance expectation for the unattributed crossings, what it is conditioned on, and what the control ensemble calibrating it can support.

##### D.1. *What the expectation is conditioned on*

One attribution set serves both sides of the comparison, the set of §5.3: the mask of Appendix C within  $\pm 50 \text{ km s}^{-1}$  in the star's own rest frame, sulphur monoxide excluded. So the 3 sulphur monoxide coincidences are unattributed crossings, and a control maximum on one of those transitions is an unattributed exceedance. A systemic velocity is measured for the stars of 277 of the census windows and the others are taken at rest, which moves the masked share by 3.0 points.

The expectation is built from the control positions, the only empirical null the data provide, and each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail — and summed over the windows 36.1 are expected to trigger by chance. The share of its own channel grid lying inside that same set is the probability that such a position would be attributed, a control maximum being free to fall on any channel. Weighted by the first that share is 20 per cent, against a median of 6.0 per cent per window, because 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers. What is left is 28.9 expected unattributed Class A crossings against 32 observed.

Both sides run over the 402 Class A windows of the primary census, so the expectation is a sum over rows and nothing is carried at an assumed rate. A further 15 Class A windows arrived with the archival tail and lie outside it, together with the two unattributed crossings toward HD 14055 they hold, which are reported beside the comparison and enter neither side of it. No window holds more than one primary crossing, so both sides count windows; the weaker cells behind those maxima are counted at their own grain in §5.3.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field, and the reserved blocks measure it in the quantity used here. Their stars reach the trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. The same blocks place the stellar rank below one half (Appendix B), which is the same departure read in the body of the distribution rather than in its tail and runs the same way. Carrying the measured ratio through gives 30.8 against the 28.9 above, and the observed 32 sits inside the range either way; the ratio is itself consistent with unity, so the uncorrected sum is the one quoted.

The 12–48 quoted beside the expectation is what 20 000

resamplings of the execution blocks allow, each window's rate redrawn from its own 512 probes: windows share blocks and are not independent trials, and the ensemble is finite. Those two terms alone give 26–32, so what widens the range is the calibration above. An added population of as many as 19 events would be absorbed by that width, and that is the limit on what the comparison can exclude.

Every Class B crossing in the survey comes from the one execution block of §D.2, and over both classes together 29.0 are expected against the same 32 observed: a coarse single pointing holds too few cells for its own field to reach the trigger by chance, and only 455 of the 1651 windows hold a control position above  $T_{\text{trig}} = 5$  at all. The Class A comparison is the survey's.

*The statistic is always the one measured over the window's whole field.* 157 windows admit integrations inside the primary beam that a narrower field would discard, recovering a median 2.1 times the on-source time and moving the statistic by a median 0.07; the whole-field value is the one adopted, on 8 rows of Table D1, and each window's control ensemble is measured with it, so neither side of the comparison mixes two fields. Where the extraction catalogue's disposition column disagrees — on 4 crossings, and the species list moves none into attribution — the ledger is the authority.

##### D.2. *Execution-block quality*

The comparison with the control positions assumes each block's ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block rather than to the one that prompted it.

*Two criteria are built from that statistic and they are not the same test.* The statistic is the median of a window's own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the median of the same quantity over the survey, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1651 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), and every one of those windows carries a crossing, attributed and unattributed alike. A *block* is judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel  $\times$  drift grid sits higher and one threshold should mean the same thing in both; a block fails when the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both. Neither was fixed in advance: the condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is that it does not depend on where the line was drawn — the failing block sits at 4.1 and the next worst at 1.12, so any threshold between them selects the same single block.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002\_Xc7fa6f\_X28a8 toward BD+05 1668. Nothing else is close, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey's entire Class B crossing population, so it carries 4 of the 56 threshold crossings — 0 of them

outranking all 512 control positions of their own window — and no Class A statistic here has ever contained it. Those 4 appear in Table D1 with their block named and enter no count elsewhere.

### D.3. *The control ensemble: its calibration, and its resolution*

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1645 census windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies — median 0.48 with each star carrying equal weight, 0.47 to 0.51 — in both array strata, the compact-array windows measured on each observation’s own antenna table rather than on a 12 m beam. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half, and Appendix B.1 compares the two samples. The expectations above are summed from the exceedance rate measured at each window’s control positions, so they do rest on that transfer and carry its measured accuracy.

*The rank is not a significance test, and the reason is structural.* Ranking the star against 512 control positions of the same field decides which windows deserve examination, but  $1/(512+1) = 1.95 \times 10^{-3}$  is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1651 windows would need a rank 64 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it, with the primary beam bounding how many a single pointing contains. That floor is the design’s central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

### D.4. *The trials budget*

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian — a budget predicting 476 windows with a control above  $T_{\text{trig}}$  against the 455 observed — while fitting each window’s effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.5 rather than 36.1, 3 per cent away.

*The on-star crossings do exceed that budget, and the attribution accounts for it.* A Gaussian tail over every searched cell predicts 37.8 stellar exceedances against the 56 found, but 18 are attributed by the set above, 4 come from the block of §D.2 and 2 fall outside the census windows, leaving 32 against the 29.0 predicted. No excess survives the attribution.

*One stratum departs from uniformity, and the departure is where the line emission is.* Counting the windows

whose stellar peak both reaches the trigger and exceeds all 512 control positions of its own window, the 402 fine-channel windows hold 12 against 0.78 expected, while the coarse stratum, three times larger and predicting 2.43, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and an unresolved carrier would appear in Class A first for the same reason. Reaching the trigger is holding a crossing, so each of those windows holds one and 10 hold a crossing the mask attributes: the departure is a statement about the windows carrying line emission and not about the fine stratum at large. Drop the trigger condition and the star leads its own ring in 2 further windows, at 4.20 and 4.36, giving 14 over all 1651 windows against 3.2 expected. One quantity, two conditions; the chance comparison uses the conditioned one.

*What is left is bounded, and the bounds are null.* The real stellar ranks agree at  $p = 0.38$  with the 220 672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

## E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has and the coverage it lacks are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3; 1 of the 1651 census windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

*The coverage of the screening is uneven.* The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window of the census, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory’s own, replayed from each block’s delivered calibration script, and was not retained window by window. *No per-channel interference flag of our own exists, and none can be recovered.* Four measurements follow, and each could have failed.

*First, clustering in sky frequency.* If interference dominated the crossing population, the crossings would pile

TABLE D1

**THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED.** ALL 56 THRESHOLD CROSSINGS, INCLUDING THE 4 IN THE FAILING BLOCK OF §5.3.2. EVERY ROW IS THE PRIMARY CROSSING OF ITS WINDOW; THE 21 WEAKER CELLS BEHIND THOSE MAXIMA ARE NOT LISTED INDIVIDUALLY, AND THE THREE OF THEM THE MASK DOES NOT ATTRIBUTE ARE DESCRIBED IN §5.3. THE COLUMNS ARE THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT AND THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC  $T_*$ , THE LARGEST STATISTIC  $T_{\text{ctrl}}$  AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET  $\Delta\nu_*$  FROM IT, THEN THE RECURRENCE TEST:  $n_{\text{rep}}$  COVERING REPEAT BLOCKS SPANNING  $\Delta t$  DAYS, THE STATISTIC  $T_{\text{pers}}$  A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC  $T_{\text{rep}}$  MEASURED THERE AT THE CARRIED DRIFT, AND THE EXCLUSION OF EQ. B1, WHICH THREE OF THESE COLUMNS REPRODUCE, TWO OF THEM ALSO GIVING THE LIMIT DEFINED IN APPENDIX B. WHERE THE EMISSION IS PRESENT AGAIN IN A COVERING BLOCK THE TEST RETURNS A RECOVERY AND NOT AN EXCLUSION, SO BOTH  $T_{\text{rep}}$  AND  $\sigma_{\text{excl}}$  ARE BLANK AND THE DISPOSITION SAYS WHICH IT WAS.  $T_*$  AND  $T_{\text{ctrl}}$  COME FROM THE SAME MEASUREMENT OF THE SAME WINDOW;  $T_{\text{ctrl}}$  GATES NOTHING AND SAYS ONLY WHICH CROSSING WAS EXAMINED FIRST, AND THE DRIFT-MAXIMISED READING OF THE CELL IS GIVEN IN TABLE 6 FOR THE TWO LEADING EVENTS AND IS NOT TABULATED FOR THE REST. ROWS WHOSE WINDOW IS DATA-QUALITY FLAGGED (§D.2) ARE MARKED, AND ON THOSE  $T_{\text{pers}}$  AND THE EXCLUSION ARE WITHHELD: BOTH ARE BUILT FROM THE DISCOVERY AMPLITUDE, AND IN A WINDOW WHOSE CONTROL RING STANDS AS HIGH AS ITS STAR THAT AMPLITUDE CANNOT BE ASSIGNED TO A POINT SOURCE AT THE STAR. 1 OF THOSE ROWS LIES OUTSIDE THE FAILING BLOCK — HD 48370, A RESOLVED DISC — AND IS WITHHELD FOR THAT REASON AND NOT FOR ITS BLOCK. A CROSSING IS ATTRIBUTED BY THE SET DEFINED ABOVE, AND A DAGGER MARKS THE 3 ROWS WHOSE COINCIDING SPECIES THAT SET EXCLUDES. THE TRANSITION NAMED IS THE NEAREST IN THE STAR'S FRAME, SO WHERE IT IS FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. BLOCK IDENTIFIERS OMIT THE CONSTANT A002\_ PREFIX.

| Star        | block           | B $\nu_{\text{topo}}$ (GHz) | $\nu_*$ (GHz) | $T_*$ $T_{\text{ctrl}}$ line                                                 | $\Delta\nu_*$ $n_{\text{rep}}$ | $\Delta t$ (d) $T_{\text{pers}}$ | $T_{\text{rep}}$ $\sigma_{\text{excl}}$ | disposition                     |
|-------------|-----------------|-----------------------------|---------------|------------------------------------------------------------------------------|--------------------------------|----------------------------------|-----------------------------------------|---------------------------------|
| BD+05 1668  | Xc7fa6f_X28a8   | 6                           | 240.1016      | 240.1054 73.398 74.69 CS(5–4)                                                | -5911.9                        | 29 0.9–113                       | n/a +0.15                               | n/a unattributed; flagged       |
| BD+05 1668  | Xc7fa6f_X28a8   | 6                           | 224.7422      | 224.7458 48.303 83.74 H <sub>2</sub> CO(3 <sub>1,2</sub> –2 <sub>1,1</sub> ) | -1264.5                        | 29 0.9–113                       | n/a +0.37                               | n/a unattributed; flagged       |
| BD+05 1668  | Xc7fa6f_X28a8   | 6                           | 226.7422      | 226.7458 41.355 75.48 CN(2 <sub>3</sub> –1 <sub>1/2</sub> )                  | +88.0                          | 29 0.9–113                       | n/a +0.66                               | n/a unattributed; flagged       |
| BD+05 1668  | Xc7fa6f_X28a8   | 6                           | 242.1953      | 242.1992 39.586 83.29 CS(5–4)                                                | -3349.2                        | 29 0.9–113                       | n/a +0.77                               | n/a unattributed; flagged       |
| $\beta$ Pic | Xf5d76d_Xeb8    | 3                           | 115.2603      | 115.2711 30.765 7.34 CO(1–0)                                                 | -0.2                           | 4 0.0–53                         | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xf5d76d_Xe19    | 3                           | 115.2603      | 115.2711 27.684 6.48 CO(1–0)                                                 | -0.2                           | 4 0.0–53                         | n/a                                     | n/a attributed; recurs          |
| HD 48370    | Xc26103_X155a   | 6                           | 230.5117      | 230.5243 26.834 18.36 CO(2–1)                                                | -17.8                          | — —                              | n/a                                     | n/a attributed; flagged         |
| $\beta$ Pic | Xf4df6f_Xc7     | 3                           | 115.2620      | 115.2711 17.907 5.76 CO(1–0)                                                 | -0.2                           | 4 0.0–53                         | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xf5d76d_Xcc5    | 3                           | 115.2604      | 115.2711 17.288 5.52 CO(1–0)                                                 | -0.2                           | 4 0.0–53                         | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xf5d76d_X32f1   | 3                           | 115.2606      | 115.2713 14.654 7.27 CO(1–0)                                                 | +0.4                           | 4 0.0–53                         | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xd9668b_X3a90   | 6                           | 230.5161      | 230.5378 11.681 9.12 CO(2–1)                                                 | -0.3                           | 5 1.0–1983                       | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xc6f1341_X1484  | 6                           | 230.5284      | 230.5379 9.832 8.20 CO(2–1)                                                  | -0.1                           | 5 25–1984                        | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | X7116f1_X1785   | 6                           | 230.5272      | 230.5384 9.677 14.13 CO(2–1)                                                 | +0.5                           | 5 25–1959                        | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xd9668b_Xa9df   | 6                           | 230.5160      | 230.5377 9.848 8.41 CO(2–1)                                                  | -0.4                           | 5 1.0–1984                       | n/a                                     | n/a attributed; recurs          |
| $\beta$ Pic | Xa7a216_X23e4   | 6                           | 230.5273      | 230.5377 7.988 6.03 CO(2–1)                                                  | -0.4                           | 5601–1313                        | n/a                                     | n/a attributed; recurs          |
| 61 Vir      | Xc079b5_X82f    | 7                           | 346.3138      | 346.3232 5.958 5.65 SO(9 <sub>8</sub> –7 <sub>7</sub> )                      | -177.6                         | 3 0.1–3.0                        | 9.7 +0.39                               | 4.9 unattributed; no recurrence |
| HD 285968   | Xd98580_X3b6f   | 6                           | 230.5021      | 230.5447 5.956 8.12 CO(2–1)                                                  | +8.7                           | 3 13–44                          | n/a                                     | n/a attributed; recurs          |
| HD 285968   | Xdb7ab7_X5b4e   | 6                           | 230.5116      | 230.5447 5.868 7.22 CO(2–1)                                                  | +8.7                           | 3 5.0–44                         | n/a                                     | n/a attributed; recurs          |
| HD 285968   | Xdb4b9a_X108f   | 6                           | 230.5100      | 230.5447 5.835 7.94 CO(2–1)                                                  | +8.7                           | 3 5.0–39                         | n/a                                     | n/a attributed; recurs          |
| CP–72 2713  | Xff0235_X4a6d   | 7                           | 344.2697      | 344.2965 5.809 5.68 SO(8 <sub>8</sub> –7 <sub>7</sub> )                      | -12.3†                         | 5 0.1–2.1                        | 6.3 -0.82                               | 4.8 unattributed; no recurrence |
| HD 139084   | B Xcd8029_Xb6b0 | 7                           | 345.8241      | 345.8241 5.763 17.70 CO(3–2)                                                 | +24.4                          | — —                              | —                                       | — attributed                    |
| HD 14055    | Xff3d8_Xd46     | 7                           | 331.7699      | 331.7664 5.575 5.99 <sup>13</sup> CO(3–2)                                    | +1068.7                        | 30 0.9–650                       | 170.4 -0.02                             | 5.6 unattributed; no recurrence |
| HD 23484    | X10b22d7_X2445  | 6                           | 230.7904      | 230.8006 5.520 5.55 CO(2–1)                                                  | +341.5                         | 4 0.1–2.1                        | 10.4 +1.25                              | 4.3 unattributed; no recurrence |
| HD 161868   | Xff45a6_X10ca8  | 7                           | 345.6338      | 345.6491 5.481 6.16 SO(2 <sub>3</sub> –2 <sub>1</sub> )                      | -48.0†                         | 13 0.0–6.1                       | 92.2 +0.24                              | 5.5 unattributed; no recurrence |
| HIP 10679   | Xa95c04_X14cd   | 6                           | 220.4185      | 220.4035 5.456 6.76 <sup>13</sup> CO(2–1)                                    | +6.5                           | — —                              | —                                       | — attributed                    |
| HD 14055    | X11adad7_X25c5e | 7                           | 345.4564      | 345.4303 5.448 6.82 SO(2 <sub>3</sub> –2 <sub>1</sub> )                      | -237.8                         | 30 0.1–653                       | 14.0 +0.03                              | 5.1 unattributed; no recurrence |
| HD 207129   | Xc19d6f_X5706   | 6                           | 230.4226      | 230.4070 5.433 5.52 CO(2–1)                                                  | -170.3                         | 4 0.8–425                        | 69.7 -0.93                              | 5.5 unattributed; no recurrence |
| $\beta$ Pic | Xd9668b_X3a90   | 6                           | 241.7512      | 241.7739 5.420 5.81 CS(5–4)                                                  | -3869.7                        | 1 1.0                            | 5.7 +0.72                               | 3.4 unattributed; no recurrence |

TABLE D2

*The crossing ledger, continued.* THE REMAINING 28 OF THE 56 THRESHOLD CROSSINGS, IN THE SAME ORDER AND WITH THE SAME COLUMNS AS TABLE D1, WHOSE CAPTION DEFINES THEM. THESE TOO ARE PRIMARY CROSSINGS, ONE TO A WINDOW.

| Star                     | block            | B $\nu_{\text{topo}}$ (GHz) | $\nu_*$ (GHz) | $T_*$ $T_{\text{ctrl}}$ line                                               | $\Delta\nu_*$ $n_{\text{rep}}$ | $\Delta t$ (d) $T_{\text{pers}}$ | $T_{\text{rep}}$ $\sigma_{\text{excl}}$ | disposition                           |
|--------------------------|------------------|-----------------------------|---------------|----------------------------------------------------------------------------|--------------------------------|----------------------------------|-----------------------------------------|---------------------------------------|
| HR 1010                  | Xc6fa08_X1053    | 6                           | 230.5133      | 230.5287 5.410 5.65 CO(2–1)                                                | -12.0                          | 7 2.0–14                         | 8.4 <+4.60                              | >2.1 attributed; not recovered        |
| ALMA J153702653-33192492 | Xff0235_X8392    | 6                           | 230.5222      | 230.5392 5.405 13.41 CO(2–1)                                               | +1.6                           | 6 18–141                         | 6.2 <+4.15                              | >1.4 attributed; not recovered        |
| HD 110411                | Xc45e29_Xb7d4    | 6                           | 219.4538      | 219.4357 5.393 5.85 C <sup>18</sup> O(2–1)                                 | -170.2                         | 1 10                             | 6.7                                     | +0.62 3.8 unattributed; no recurrence |
| HD 31392                 | X129754b_X6946   | 6                           | 231.1998      | 231.2231 5.289 5.81 H(30a)                                                 | -876.3                         | 8 0.1–41                         | 15.7                                    | -0.26 5.1 unattributed; no recurrence |
| HD 14055                 | Xff99e1_X24be0   | 7                           | 331.7078      | 331.7035 5.277 5.69 <sup>13</sup> CO(3–2)                                  | +1011.6                        | 30 0.9–652                       | 252.2                                   | +0.15 5.3 unattributed; no recurrence |
| AU Mic                   | Xa42f75_X319     | 6                           | 230.6864      | 230.6705 5.274 6.07 CO(2–1)                                                | +172.3                         | 2310–455                         | 6.4                                     | -0.16 4.2 unattributed; no recurrence |
| HD 69830                 | X12209f7_X16deb7 | 7                           | 322.4325      | 322.4444 5.259 5.42 C <sup>18</sup> O(3–2)                                 | -6268.5                        | 5 0.1–29                         | 8.9                                     | -0.48 4.8 unattributed; no recurrence |
| ALMA J153702653-33192492 | Xfe62c1_X398     | 6                           | 216.1844      | 216.2036 5.228 6.02 SiO(5–4)                                               | -1244.6                        | 7 18–126                         | 11.0                                    | +1.03 4.3 unattributed; no recurrence |
| AU Mic                   | X899169_X1838    | 6                           | 230.8252      | 230.8294 5.221 5.85 CO(2–1)                                                | +379.0                         | 2145–310                         | 10.0                                    | +0.19 4.5 unattributed; no recurrence |
| HD 14055                 | Xff99e1_X2c3bb   | 7                           | 345.5256      | 345.5216 5.186 5.66 SO(2 <sub>3</sub> –2 <sub>1</sub> )                    | -158.6                         | 30 0.1–651                       | 172.8                                   | +2.00 5.1 unattributed; no recurrence |
| HD 14055                 | Xffde1_X3835     | 7                           | 345.0099      | 345.0070 5.177 5.66 SO(2 <sub>3</sub> –2 <sub>1</sub> )                    | -604.8                         | 30 0.1–649                       | 187.2                                   | -1.78 5.2 unattributed; no recurrence |
| HD 14055                 | Xff99e1_X2b2ee   | 7                           | 344.7007      | 344.6965 5.175 5.92 SO(8 <sub>8</sub> –7 <sub>7</sub> )                    | +336.0                         | 30 0.1–651                       | 210.7                                   | -0.00 5.2 unattributed; no recurrence |
| LHS 1140                 | Xd9d7c7_X6fd2    | 6                           | 230.6355      | 230.6268 5.136 5.68 CO(2–1)                                                | +115.5                         | 4 5.0–27                         | 7.3                                     | +0.33 4.0 unattributed; no recurrence |
| HD 10647                 | Xff294f_Xaf70    | 7                           | 331.6570      | 331.6943 5.120 5.91 <sup>13</sup> CO(3–2)                                  | +1003.3                        | 10 0.1–592                       | 125.8                                   | +0.39 5.1 unattributed; no recurrence |
| CD–57 1054               | Xc5f25f_X5aa4    | 7                           | 346.5099      | 346.5265 5.108 5.71 SO(9 <sub>8</sub> –7 <sub>7</sub> )                    | -1.7†                          | 1 39                             | 6.0                                     | -0.27 4.1 unattributed; no recurrence |
| ALMA J153702653-33192492 | Xf8f6a9_X10359   | 6                           | 217.7292      | 217.7269 5.091 5.41 H <sub>2</sub> CO(3 <sub>0,3</sub> –2 <sub>0,2</sub> ) | -680.5                         | 7 3.8–242                        | 11.6                                    | +0.07 4.6 unattributed; no recurrence |
| GJ 849                   | Xd9668b_X841a    | 6                           | 230.8769      | 230.8581 5.090 5.83 CO(2–1)                                                | +416.3                         | 5 0.1–51                         | 9.3                                     | -1.69 5.3 unattributed; no recurrence |
| ALMA J153702653-33192492 | Xfe62c1_X398     | 6                           | 218.8729      | 218.8923 5.090 5.56 H <sub>2</sub> CO(3 <sub>2,1</sub> –2 <sub>2,0</sub> ) | +181.2                         | 7 18–126                         | 10.5                                    | -1.04 5.0 unattributed; no recurrence |
| HD 14055                 | X101fdc_X29ae    | 7                           | 331.1559      | 331.1546 5.082 6.26 <sup>13</sup> CO(3–2)                                  | +513.8                         | — —                              | —                                       | — unattributed; no spectrum           |
| $\beta$ Pic              | Xd9668b_Xa9df    | 6                           | 241.8367      | 241.8594 5.079 5.99 CS(5–4)                                                | -3765.1                        | 1 1.0                            | 4.8                                     | -0.63 3.9 unattributed; no recurrence |
| ALMA J153702653-33192492 | Xf8f6a9_X10359   | 6                           | 218.6711      | 218.6688 5.048 5.77 H <sub>2</sub> CO(3 <sub>2,1</sub> –2 <sub>2,0</sub> ) | -125.1                         | 7 3.8–242                        | 11.6                                    | +0.00 4.6 unattributed; no recurrence |
| HD 69830                 | X12277ed_X178c6  | 7                           | 321.2005      | 321.2149 5.046 6.13 C <sup>18</sup> O(3–2)                                 | -7387.7                        | 5 6.0–36                         | 10.2                                    | -0.46 4.7 unattributed; no recurrence |
| HR 1010                  | Xc66ce1_Xaa9     | 6                           | 230.8348      | 230.8491 5.044 5.72 CO(2–1)                                                | +404.6                         | 7 1.1–14                         | 10.3                                    | -1.53 5.2 unattributed; no recurrence |
| 61 Vir                   | Xc067f7_X822     | 7                           | 345.5599      | 345.5678 5.038 5.79 SO(2 <sub>3</sub> –2 <sub>1</sub> )                    | -118.5                         | 3 2.0–3.0                        | 9.3                                     | +0.45 4.2 unattributed; no recurrence |
| HD 170773                | Xff0235_X8c42    | 7                           | 331.0101      | 331.0235 5.029 5.75 <sup>13</sup> CO(3–2)                                  | +394.8                         | 4 0.9–113                        | 74.8                                    | +1.56 4.9 unattributed; no recurrence |
| ALMA J153702653-33192492 | Xd9d7c7_Xbb64    | 7                           | 347.2179      | 347.1890 5.009 5.68 SiO(8–7)                                               | -122.2                         | 1 0.1                            | 5.0                                     | -2.06 5.0 unattributed; no recurrence |
| HD 14055                 | X1190de8_Xd526   | 7                           | 344.4676      | 344.4515 5.006 5.96 SO(8 <sub>8</sub> –7 <sub>7</sub> )                    | +122.7                         | 30 20–611                        | 11.6                                    | -0.60 4.8 unattributed; no recurrence |
| $\eta$ Crv               | X122b6ff_X1041e  | 7                           | 357.2815      | 357.2475 5.006 5.50 HCO <sup>+</sup> (4–3)                                 | +431.3                         | 1 87                             | 11.0                                    | -0.62 4.8 unattributed; no recurrence |

up at particular sky frequencies. They do cluster — but on the survey’s own molecular lines. The test needs a measured peak frequency in the frame an interfering carrier would stand still in, so it runs topocentrically over the 51 of the 56 crossings for which one was measured, against the transitions the search was tuned to. The other 5 are carbon monoxide toward  $\beta$  Pic, four in CO(1–0) and one in CO(2–1) of its ten such crossings; for those the catalogue carries only the velocity offset from the line, so the frequency Table D1 prints is recovered from that offset rather than measured, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of the 51, 11 fall within  $\pm 50 \text{ km s}^{-1}$  of a searched transition against 4.7 expected if each window’s peak fell anywhere on that window’s own channel grid,  $p = 0.0007$ . The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2→1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ( $p = 0.111$ ). *The concentration is the attribution, not a residue left over after it.* Table D1 counts the same band as 14 crossings with 11 attributed, over a different set — all 56 of them, each attributed in its own star’s rest frame — and the two rows it adds are the ones this test cannot hold: one whose peak frequency the release records only as an offset from the line and one whose window the release does not record as a crossing at all.

*Second, the same test against the atmosphere.* A residual of the atmospheric calibration stands still in topocentric frequency exactly as a transmitter does, and it is narrow where the atmosphere has lines. 570 transitions of ozone, water and oxygen lie within 89.6–873.1 GHz (Pickett et al. 1998), a median 0.9 GHz apart. No crossing of either sample falls within 25 MHz of one; the median separation is 0.81 GHz; and 1 crossing lies inside the same  $\pm 50 \text{ km s}^{-1}$  window used above, against 2.1 expected from the windows’ own channel grids. The one concentration a reader will notice — 15 crossings of the census and the hold-out between 344.2 and 346.4 GHz — is not atmospheric either, because that interval holds 0 telluric transitions; 12 of the 43 unattributed crossings whose window the catalogue tabulates fall in it against  $11.8 \pm 0.9$  expected from where the spectral windows were tuned. *It is the carbon monoxide tuning, seen from the other side.*

*Third, fixed intermediate frequency.* An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks’ fractional positions within their own windows have median 0.50 and are uniform.

*Fourth, a cross-target occupancy screen over every window the pipeline extracted, not only the crossings.* That set is wider than the census: 3027 windows toward 133 stars, being the census together with the rows its audit removed, the reserved hold-out, the archival extension, the out-of-sample control, and 64 windows in 16 blocks that entered none of them. Interference belongs to the observatory and not to the star, so a terrestrial carrier

should recur at the same sky frequency toward *unrelated* stars. It does not, and the sharpest place to look is the two unattributed crossings that outrank every control position in their own window, 61 Vir and CP–72 2713, both of which carry a measured frequency. Taking them in that order, 346.314 and 344.270 GHz are covered by 36 and 42 further windows, toward 24 and 8 other stars in 32 and 42 other blocks, and none of them carries a crossing in the same channel. With each window’s peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against  $554 \pm 13$  expected ( $p = 0.15$ ); among the 109 windows whose peak exceeds the trigger it is 4 against  $5.2 \pm 1.9$ , i.e. *below* the null, and with the molecular-line mask applied 1 against  $2.1 \pm 1.3$ . There is no excess to explain. The mask has to be applied first, because 35 of those 109 windows sit on a masked transition (CO(2–1)  $\times 17$ , CO(1–0)  $\times 10$ , CO(3–2)  $\times 4$ ,  $^{13}\text{CO}(2–1) \times 3$ , SiO(5–4)  $\times 1$ ), and a carbon monoxide line falls at nearly the same sky frequency toward every nearby star — exactly the coincidence the screen is looking for.

*One execution block fails these tests, and it fails them as a block rather than as a carrier.* Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside  $\beta$  Pictoris and HD 48370, and §5.3.2 identifies the failure. Whether interference accounts for it is what belongs here, and three readings say not: the visibility fit returns a large imaginary part, placing the emission away from the phase centre; the block’s 24 sibling executions of the same field, array and spectral setup produce no crossing at all; and each of the 4 frequencies is covered by 127–137 further windows toward 17–22 other stars, of which 0 carries a crossing in the same channel.

*One reading does point at a mechanism, and it can be tested across the survey.* 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency; and a feature with no geometric fringe phase — a spur common to every antenna — does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–8.1 arcsec from the phase centre in 75 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against probes at the same distance from the star but elsewhere in azimuth — the radius has to be matched, or a centrally concentrated disc would imitate the effect — those probes are higher by  $+0.09 \pm 0.08$  in the statistic, consistent with zero and bounded by 0.26, smaller by a factor of about 33 than the 8.4 by which this block’s ring is raised. *The phase centre is not where the excess lives.* The evidence supports a data-quality failure of one execution block, and the block criterion of Appendix D.2 — which fails 1 block of the 404 searched — is what disposes of these 4 crossings (§5.3.2).

*None of this substitutes for the ON/OFF cadence a dedicated search uses, and it is not presented as one.* The claim is only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy

cannot do better. The four tests establish that the crossing population carries none of the signatures interference

leaves — a statement about the population and not a clearance of any individual crossing.

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